

SpectroElectroChem Suite

USER MANUAL

Version 6.3.0

Open-source software for Raman, SERS, absorption and fluorescence spectroelectrochemistry, RRDE, ECL, diffusion, EIS, stripping-voltammetry and electrochemical surface-activation/SERS analysis

Author	Prof. Dr. Achim Habekost
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Repository	https://github.com/Achim-Habekost/SpectroElectroChem-Suite
License	MIT License

Parts of the source code were developed with the assistance from OpenAI ChatGPT and were subsequently reviewed, scientifically validated, modified and substantially extended by the author.

Version 6.3.0 — Update and release notes

This update accompanies the existing 64-page illustrated user manual. All original figures and screenshots are retained. The original chapter text documents the established workflows; where a version-specific detail differs, the notes below take precedence.

Installation and distribution

Windows users can choose the v6.3.0 installer **SpectroElectroChem_Suite_Setup_v6_3_0.exe** or the packaged executable **SpectroElectroChem_Suite_v6_3_0.exe**. The standalone Windows build does not require a separate Python installation. The application launcher starts individual analysis modules from the packaged executable. The installer and executable are separate release assets; do not confuse them with the small source-code ZIP.

Module 2 — Raman/SERS voltammograms

Potential-dependent data can follow a forward scan and a reverse scan. The first matrix row must contain the actual electrode potentials in measurement order. Repeated potentials on the return path represent different measurement times and must not be removed merely because their numerical values are identical.

Module 3 — Absorpto-/fluorovoltammograms

The forward and reverse potential sweeps are considered in potential-resolved evaluation. Keep the chronological sequence of spectra and the actual potential values in the first row; use the first column for wavelength. In reflection-based experiments, interpret apparent absorbance or relative reflectance changes according to the chosen reference rather than assuming transmission absorbance.

Module 5 — Spectro-CV synchronization

Synchronization and evaluation of forward/reverse scans have been refined. The potential derivative dA/dE must use the actual signed potential increment; inspect the turning points before interpretation. The module provides current-unit selection (A, mA or µA), spectral plotting, and wavelength-interval integration using adjustable boundaries. Verify timing, scan rate, and cycle count against the experimental protocol.

Reproducibility and validation

Preserve original CSV data, instrument settings, electrode reference, scan direction, spectral integration time, and software version. Inspect all plotted results and exports before drawing scientific conclusions. The current software and module interfaces take precedence over any older screenshot in the original illustrated manual.

Persistent links

Repository: https://github.com/Achim-Habekost/SpectroElectroChem-Suite
Version 6.3.0 archive DOI: https://doi.org/10.5281/zenodo.23664353
All-version DOI: https://doi.org/10.5281/zenodo.21283231

Document status and scope

This manual describes SpectroElectroChem Suite version 6.0. It is intended for scientists, students, and technical staff who wish to analyze their own comma-separated value (CSV) files without modifying the Python source code. The software provides graphical workflows for Raman, SERS, absorption and fluorescence spectroelectrochemistry, RRDE analysis, Spectro-CV and ECL synchronization, diffusion-coefficient determination, electrochemical impedance spectroscopy (EIS) and stripping voltammetry, including signal processing, fitting, visualization and numerical export.

> Important: SpectroElectroChem Suite is available in two forms. (1) A ready-to-run Windows program: a self-contained executable (SpectroElectroChem_Suite.exe) that can be started directly, plus an optional Windows installer that adds Start-menu and desktop shortcuts. Neither requires a separate Python installation. (2) The Python source distribution, which is started with the included batch files and requires Python 3.10 or newer. Both forms provide the same ten analysis modules: Raman Spectrum Analysis; SERS/Raman Voltammogram; Absorpto-/Fluorovoltammogram; RRDE Analysis; Spectro-CV Synchronization; ECL Synchronization (Integrated Signal); Diffusion Coefficient Analysis; EIS Analysis; Stripping Voltammetry; and Electrochemical Surface Activation & SERS Analysis.

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1. Overview

1.1 Purpose

SpectroElectroChem Suite is a modular desktop application for processing, visualizing and exporting spectroscopic and spectro-electrochemical data. It is designed to reduce the amount of manual spreadsheet work required for multi-spectrum data sets while preserving transparent numerical outputs for independent inspection.

1.2 Supported analytical workflows

- Single Raman spectra or multiple Raman spectra stored in columns: smoothing, baseline correction, peak analysis and waterfall visualization.
- Raman and SERS voltammograms: potential-resolved Raman intensity matrices.
- Absorptovoltammograms: potential-resolved absorption spectra.
- Fluorovoltammograms: potential-resolved fluorescence spectra.
- RRDE voltammograms: rotation-rate-resolved disk and ring currents for the rotating ring–disk electrode.
- Spectro-CV Synchronization: synchronization of cyclic voltammetry with time-resolved absorption, fluorescence/ECL or Raman spectra; potential assignment, wavelength/peak traces, spectral derivatives and multi-cycle evaluation.
- ECL Synchronization (Integrated Signal): synchronization of CV and non-wavelength-resolved ECL signals, cycle averaging, signal integration, co-reactant comparison and concentration-series calibration.

- Diffusion Coefficient Analysis: determination of diffusion coefficients by Cottrell, Anson and Randles–Ševčík analysis, complemented by Nicholson electron-transfer kinetics and scan-rate diagnostics.
- EIS Analysis: Nyquist and Bode visualization, Randles-type equivalent-circuit fitting, residual analysis and Kramers–Kronig consistency checking.

1.3 Principal outputs

- Processed Excel workbooks containing numerical results and metadata.
- Static PNG and PDF figures for reports and publications.
- Interactive HTML figures for rotation, zooming and visual inspection.
- 3D surfaces, heatmaps, contour plots and waterfall diagrams.
- Raw, smoothed, baseline-corrected and vertically offset data, depending on the selected module.
- Levich, Koutecký–Levich, Tafel and Butler–Volmer analyses.
- Spectro-CV Synchronization: synchronized CV/spectral data, selected wavelength or Raman-shift traces, potential-dependent derivatives, static PNG figures, interactive 3D waterfall plots and Excel exports.
- ECL Synchronization (Integrated Signal): synchronized CV/ECL traces, individual and mean cycles, integrated ECL intensities, calibration plots, unknown-concentration evaluation and Excel/PNG exports.
- Diffusion Coefficient Analysis: Cottrell, Anson and Randles–Ševčík plots with fitted diffusion coefficients, Nicholson kinetic parameters, scan-rate diagnostics and numerical Excel output.
- EIS Analysis: Nyquist and Bode plots, fitted equivalent-circuit parameters, residual plots, Kramers–Kronig consistency results and Excel export of experimental and fitted data.

1.4 Intended use and limitations

The software is intended as a scientific analysis and visualization aid. It does not replace validation of instrumental data, inspection of raw spectra, calibration, uncertainty analysis or expert interpretation. Parameters such as the smoothing window, baseline model, peak prominence and waterfall offset must be selected with reference to the spectral resolution, the signal-to-noise ratio and the specific scientific question.

2. Obtaining the software

2.1 Zenodo archive

The archived version can be accessed through the persistent DOI:

https://doi.org/10.5281/zenodo.21283231

On the Zenodo record page, download the ZIP archive associated with version 6.0. Zenodo provides a permanently archived snapshot together with citation metadata.

2.2 GitHub repository

The development repository and release page are available at:

https://github.com/Achim-Habekost/SpectroElectroChem-Suite

Use GitHub for source inspection, issue reporting and future releases. Use the Zenodo DOI when citing the archived version used in a scientific study.

2.3 After downloading

- Save the downloaded ZIP archive to a local folder.
- Right-click the archive and choose Extract All.
- Open the extracted project folder. Do not run batch files directly from inside the ZIP archive.

> Security notice: Windows may mark downloaded batch files as downloaded from the Internet. Inspect the files and source code before execution, and use only the official GitHub or Zenodo release.

3. System requirements

Component	Minimum / required	Recommended
Operating system	Windows 10 or 11	64-bit Windows 11
Python	Python 3.10 or newer	Current stable 64-bit Python
Memory	4 GB RAM	8 GB or more for large matrices
Storage	Approx. 1 GB plus output files	Several GB for large Excel/HTML exports
Internet	Needed initially for package installation	Needed to load Plotly from a CDN in some HTML outputs
Spreadsheet viewer	Optional	Microsoft Excel or LibreOffice Calc
Web browser	Modern browser	Firefox, Edge or Chrome

3.1 Required Python packages

The release requirements file lists the packages used by the different modules. The principal packages are:

- NumPy – numerical arrays and matrix operations.
- pandas – CSV input and tabular data handling.
- SciPy – Savitzky–Golay filtering, peak processing and numerical routines.
- pybaselines – baseline-estimation algorithms, where used.
- Matplotlib – static plots and PDF/PNG export.
- Plotly – interactive HTML visualizations.
- openpyxl – Excel workbook creation.
- PySide6 and/or Tkinter – graphical user interfaces, depending on the module implementation.

4. Installation on Windows

Revision (v6.0): The self-contained Windows program — the ready-to-run executable and the optional installer described in Section 4.5 — is the recommended installation route and requires no separate Python installation. Installation from the Python source distribution (Sections 4.1–4.4) is provided as an alternative for advanced users and developers and is summarised in Appendix E.

4.1 Install Python

Alternative (advanced) route: the following Python-source steps (Sections 4.1–4.4) are only required if you do not use the executable or installer from Section 4.5. A condensed version is given in Appendix E.

- Download Python 3.10 or newer from the official Python website.
- During installation, enable Add Python to PATH if that option is shown.
- Complete the installation and restart Windows, or at least reopen File Explorer and Command Prompt.

4.2 Install required packages

In the extracted program folder, double-click:

```
Install_required_Python_packages.bat
```

A console window will open and install the packages listed in requirements.txt. Internet access is normally required during this step. The step is generally needed only once for a given Python environment.

4.3 Test the installation

Run:

```
Test_Installation.bat
```

The test verifies that Python and the principal dependencies can be imported. If the console window closes immediately, launch the batch file from Command Prompt to retain the error message.

4.4 Manual installation alternative

Alternatively, open Command Prompt in the program folder and run:

```
py -m pip install -r requirements.txt
```

If the py launcher is unavailable, try:

```
python -m pip install -r requirements.txt
```

> Multiple Python installations: If package installation succeeds but the application still reports missing modules, the batch files may be using a different Python installation from the one in which the packages were installed. Use “py -0p” in Command Prompt to list the installed Python interpreters.

4.5 Installation using the Windows executable or installer

From version 6.0, the suite is also distributed as a ready-to-run Windows program that does not require a separate Python installation. Two variants are provided.

Portable version. Extract the distributed program folder and double-click the executable:

```
SpectroElectroChem_Suite.exe
```

All required libraries are bundled inside the folder. Keep the executable together with the accompanying files and sub-folders; do not move the .exe out of its folder.

Windows installer (setup). Alternatively, run the setup program (for example SpectroElectroChem_Suite_Setup.exe). The installer copies the program into the standard applications folder and creates Start-menu and desktop shortcuts. The application can then be started like any other installed Windows program.

The executable and the installer are built from the same source code as the Python distribution and provide the same analysis modules. No Python interpreter, batch file or package installation is needed for these variants.

Security notice: Because the executable is newly published and not code-signed, Windows SmartScreen or the antivirus software may display a warning on first start. Verify that the file was obtained from the official GitHub or Zenodo release, then choose “More info” and “Run anyway” if you trust the source.

5. Starting and closing the application

5.1 Standard start

Double-click:

```
Start_SpectroElectroChem_Suite.bat
```

The launcher adds the src folder to the Python search path and opens the central application window. Depending on the release build, the main window offers module buttons or menu entries for Raman spectrum analysis, Raman/SERS voltammograms, and absorption/fluorescence voltammograms, RRDE, CV-Spectra Synchronization, ECL, Diffusion, EIS, Stripping Voltammetry and Electrochemical Surface Activation & SERS Analysis.

5.2 Start from Command Prompt

```
cd "C:\path\to\SpectroElectroChem_Suite"
python main.py
```

5.3 Starting from the Windows executable

If the ready-to-run program or the installed version is used, start the suite directly:

- Portable version: double-click SpectroElectroChem_Suite.exe inside the program folder.
- Installed version: use the Start-menu entry or the desktop shortcut created by the installer.

The central application window opens with buttons or menu entries for the available analysis modules. No Command Prompt, batch file or Python installation is required in this case.

5.4 Closing

Close module windows using their normal Close or X button. Allow ongoing Excel or HTML export operations to finish before closing. Interactive HTML plots remain available as independent files after the application has been closed.

6. Input-file specifications

6.1 General CSV rules

- Use a true CSV text file, not an .xlsx file merely renamed to .csv.
- Decimal points and decimal commas are accepted by many import paths, but a decimal point is recommended for portability.
- Do not mix text comments into the numerical matrix unless the module explicitly supports metadata rows.
- Avoid merged cells, hidden rows and formulas; export numerical values only.
- Remove completely empty trailing rows and columns where practical.
- Keep a copy of the unmodified raw instrument export.

6.2 Matrix format for Raman/SERS voltammograms

The first row contains the potentials, the first column contains the Raman shifts (wavenumbers), and all remaining cells contain Raman intensities.

	−0.40	−0.30	−0.20	−0.10
300	120	125	131	140
305	118	124	130	139
310	119	126	134	143
315	123	129	138	148

6.3 Matrix format for absorption and fluorescence voltammograms

The first row contains the potentials and the first column contains the wavelengths. The remaining cells contain absorbance, fluorescence intensity or another measured optical response. If there is only one absorbance or fluorescence spectrum, the potential row is missing.

	0.00	0.10	0.20	0.30
400	0.021	0.028	0.039	0.051
405	0.023	0.031	0.043	0.056
410	0.026	0.035	0.048	0.062
415	0.030	0.040	0.054	0.069

6.4 Raman spectrum table format

For Raman spectrum analysis, the first numerical column contains the Raman shift and the remaining columns contain one or more intensity spectra. Header labels may be used to identify individual spectra at various potentials. The exact import options are displayed by the module.

6.5 RRDE

RRDE data are imported from a CSV table in which the first column contains the electrode potential. The following columns contain alternating disk and ring currents for the individual rotation rates (Disk 1, Ring 1, Disk 2, Ring 2, ...). The first row identifies the corresponding rotation speeds. For background correction with a separate N_2 measurement, the background file should contain the same rotation rates and corresponding disk/ring current pairs. Slight differences in the potential grids are handled by interpolation during the correction procedure.

6.6 Spectro-CV Synchronization

Two CSV files are required: a CV file and a spectral file. In the CV file, the first column contains the potential and the following column(s) contain the current data. The spectral file contains the spectral coordinate (wavelength for absorption/fluorescence or Raman shift for Raman measurements) and the spectral intensity matrix. The original non-transposed arrangement, with the spectral coordinate in the first column, is accepted. The number of spectra may differ from the number of electrochemical data points because the two data sets are synchronized numerically. Correct entry of the experimental sequence, potential limits, scan direction and number of CV cycles is essential.

6.7 ECL Synchronization (Integrated Signal)

The ECL module accepts electrochemical and integrated-light data arranged as separate blocks in the supported CSV layout. The electrochemical block contains potential and current traces for the recorded cycles; the ECL block contains its own potential- or time-related axis and the corresponding ECL intensity traces. The two sampling grids do not have to be identical. For repeated measurements, specify the number of cycles. For concentration-series experiments, also specify the number of concentrations and their values in mol/L; scientific notation such as 1e-6 is accepted. Data sets recorded with and without co-reactant, and an optional unknown sample, must use the same acquisition conditions when they are to be compared quantitatively.

6.8 Diffusion Coefficient Analysis

The diffusion module uses method-specific CSV files. For Cottrell analysis, provide chronoamperometric data containing time and current. For Anson analysis, provide chronocoulometric data containing time and charge. For Randles–Ševčík analysis, provide cyclic voltammograms recorded at several scan rates, with potential and the corresponding current traces. Enter the concentration c (mol/L), electron number n, electrode area A (cm²) and temperature T (K), together with the scan rates where required. Consistent units are essential; the program reports D in cm²/s when the input quantities are supplied in the specified centimetre-based unit system.

6.9 EIS Analysis

EIS input consists of frequency-dependent impedance data. The CSV file must contain the frequency and the corresponding real and imaginary impedance components, or equivalent Nyquist/Bode columns supported by the module. Use the column-mapping controls to assign the imported quantities correctly before fitting. Preserve the original frequency values and units from the instrument export. Nyquist and Bode data should describe the same measurement when they are analyzed together. Inspect the preview before selecting and fitting a Randles-type equivalent circuit.

6.10 Stripping Voltammetry

The Stripping Voltammetry module accepts separate CSV files for square-wave voltammetry (SWV) and differential-pulse voltammetry (DPV). In each calibration file, the first column contains the electrode potential and the first row contains the concentrations of the corresponding voltammograms. The remaining columns contain the measured currents. SWV and DPV data can be loaded and evaluated together. An additional CSV file for a sample of unknown concentration can be loaded when quantitative determination is required. Use identical units and comparable measurement conditions for standards and unknown samples.

6.11 Electrochemical Surface Activation & SERS Analysis

For each metal (Au and/or Ag), load a Raman spectrum recorded before activation and a corresponding spectrum recorded after electrochemical activation. The spectral x-axis may be supplied either directly as Raman shift (cm^{-1}) or as wavelength (nm). For wavelength input, enter the laser wavelength; the program converts the measured wavelength automatically to Raman shift. The current default laser wavelength is 785 nm and can be changed by the user. Optional cyclic voltammograms can be loaded separately for Au before/after and Ag before/after activation. Comparative measurements should use identical optical acquisition conditions whenever quantitative enhancement ratios are interpreted.

6.12 Axis order

Some modules sort the potential and spectral axes automatically. Nevertheless, ascending, unique numerical axis values are recommended. Duplicate potentials or duplicate wavelengths/wavenumbers should be resolved before analysis unless they represent intentional replicate measurements.

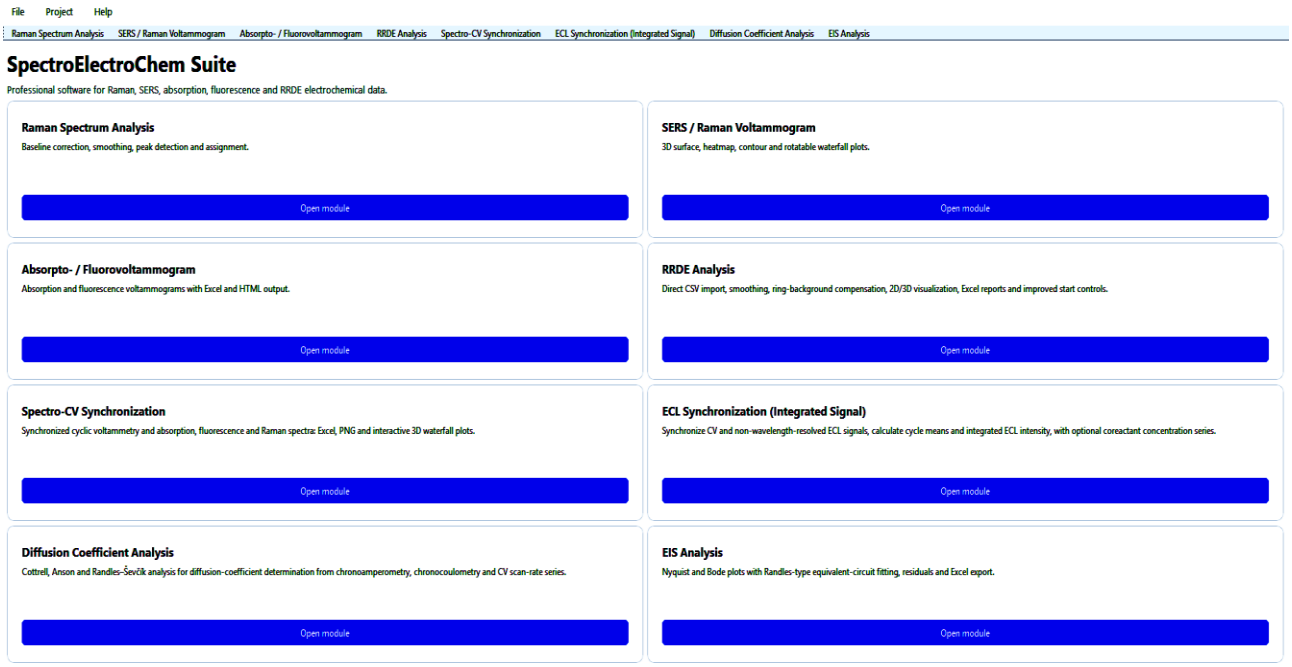


Figure 1: Initial input

7. Raman spectrum analysis

7.1 Purpose

This module processes individual Raman spectra or collections of spectra stored in columns. It can generate raw, smoothed, baseline-corrected and combined data sets, locate peaks, calculate peak parameters and create waterfall plots.

7.2 Typical workflow

- Open the Raman Spectrum Analysis module.
- Select the CSV input file.
- Enter or confirm the Raman shift range.
- Choose the baseline correction parameters.
- Choose the Savitzky–Golay smoothing parameters.
- Set the peak-finding parameters, if peak analysis is required.
- Select an optional reference peak list, if identification is required.
- Set the maximum Raman intensity for plotting, if a fixed scale is desired.
- Set the waterfall vertical offset and plot orientation, where available.
- Run the analysis and inspect the generated control plots before accepting the results.

7.3 Baseline correction

Baseline correction estimates the slowly varying background and subtracts it from the measured spectrum. A visually smooth baseline-corrected spectrum is not sufficient evidence that the baseline is physically appropriate. Depending on the current plugin, the module may provide polynomial or penalized-baseline methods. Baseline correction should remove broad fluorescence or instrumental background (e.g. scattering) without distorting genuine broad Raman bands.

> Scientific check: Always inspect the baseline overlay. A visually smooth baseline-corrected spectrum is not sufficient evidence that the baseline is physically appropriate. Check low-intensity regions, broad bands and the spectrum edges.

7.4 Savitzky–Golay smoothing

Savitzky–Golay smoothing performs a local polynomial fit. The window length must be odd and larger than the polynomial order. A window that is too large can reduce peak heights, merge neighbouring bands or alter the FWHM. Select a window consistent with the sampling interval and the expected peak width.

7.5 Peak analysis

Peak finding typically uses prominence and minimum-distance criteria. The output may include peak position, intensity and full width at half maximum (FWHM). Peak identification against a reference list is tolerance-based and must be treated as a tentative assignment, not as definitive chemical proof.

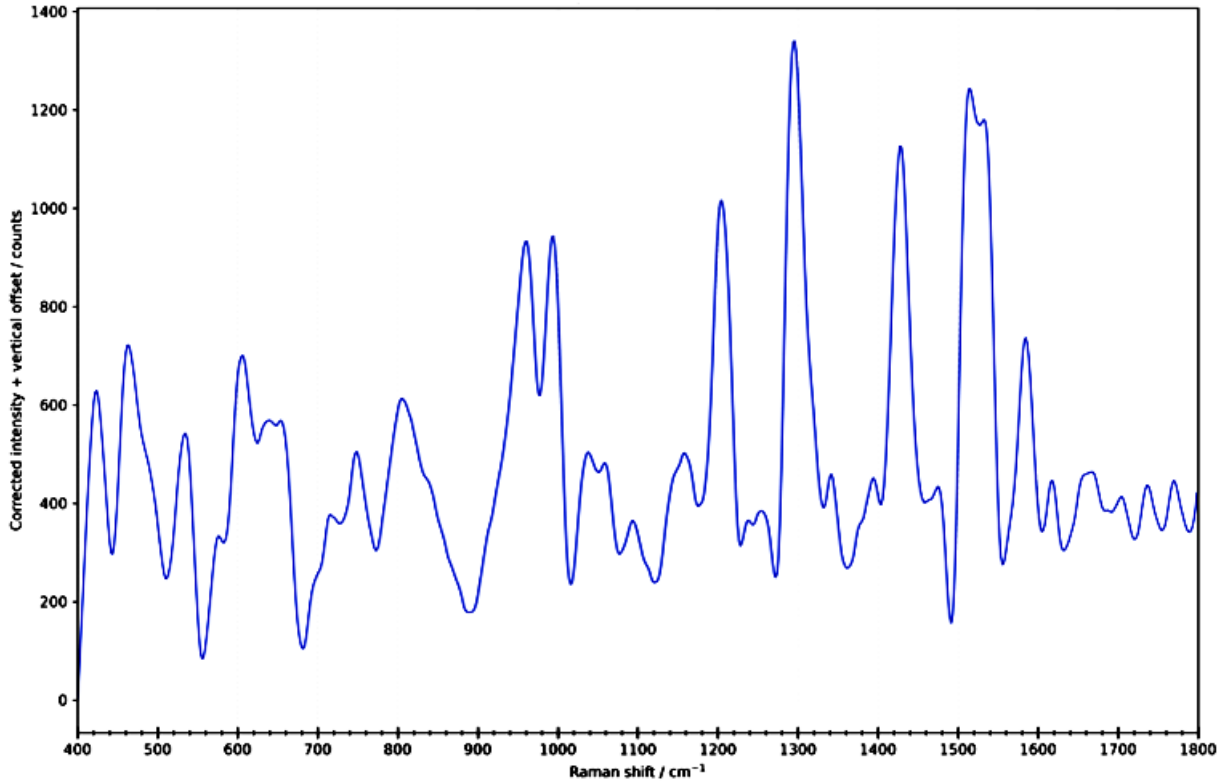


Figure 2: Smoothed and baseline-corrected Raman spectrum (rubrene).

7.6 Raman Excel workbook

Depending on the software version the Excel workbook typically contains the following worksheets:

- Original spectra
- Smoothed-only spectra
- Baseline-corrected-only spectra
- Baseline-corrected and smoothed spectra
- Calculated baselines
- 2D waterfall data
- Peak parameters
- Peak identification results
- Reference data used
- Summary/interpretation

8. Raman/SERS voltammograms

8.1 Purpose

The Raman/SERS voltammogram module processes a spectral intensity matrix as a function of Raman shift and electrode potential. It is suitable for potential-dependent Raman or surface-enhanced Raman data.

8.2 Loading and range selection

- Select the CSV matrix file.
- Select an output folder, or leave the output-folder field empty to use the input-file folder.
- Use the range-reading function, if available, to display the detected Raman-shift and potential ranges.
- Enter the desired lower and upper Raman-shift limits.

8.3 Smoothing options

- None – preserves the measured matrix.
- Moving average – simple smoothing over a selected number of points.
- Savitzky–Golay – polynomial smoothing with a selected odd window length and polynomial order.

8.4 Intensity scaling

- Raw / linear – preserves the original plotted magnitude.
- Clip at the 99th percentile + log1p reduces the influence of extreme spikes on the color scale.
- log1p – compresses the dynamic range after shifting negative values, if necessary.
- Clip99 + log1p – combines spike suppression and logarithmic compression.

> Interpretation: Scaling modes modify the visualization matrix and may also be exported as plotted values. They do not convert arbitrary intensity units into quantitative concentrations.

8.5 Baseline correction and peak parameters

- SERS spectra can exhibit substantial fluorescence or other slowly varying backgrounds; whether baseline correction is required must therefore be decided from the measured spectrum rather than from the use of SERS itself (see Figure 2).

Peak No	Position (cm^{-1})	Intensity	FWHM (cm^{-1})
1	1769.309	137.2264	12.64709
2	1736.404	168.6106	14.29673
3	1703.5	179.6649	8.804291

4	1666.939	254.3467	33.94823
5	1617.582	232.0855	11.76216
6	1584.677	512.9232	22.00155
7	1515.211	1014.986	41.86721
8	1427.465	977.9257	27.86155
9	1394.561	281.1656	8.65824
10	1341.547	222.2908	9.463366
11	1295.846	1068.63	28.16962
12	1253.801	116.1348	28.37035
13	1204.444	790.3948	26.97438
14	1158.743	273.9904	21.33758
15	1092.934	118.8374	11.15744
16	1036.264	360.092	39.3444
17	994.219	844.1082	69.27292
18	961.315	816.1149	19.09671
19	804.103	596.9451	68.56732
20	749.262	444.6386	18.29964
21	635.923	543.9664	16.40546
22	606.674	676.6498	75.71277
23	533.553	474.0955	19.23679
24	464.087	735.6446	52.75802
25	423.87	594.0613	18.36885

8.6 Manual intensity limits

When automatic intensity scaling is disabled, enter minimum and maximum intensity values. Interactive 3D figures may also include buttons for automatic Z scaling, manual Z scaling and percentile-based ranges.

8.7 Waterfall and surface plots

The waterfall plot displays each spectrum with a vertical displacement. Use the vertical offset to prevent overlap while preserving relative peak patterns. Interactive plots can be rotated; static plots are intended for publications and reports. Avoid offsets so large that potential-dependent trends become visually exaggerated.

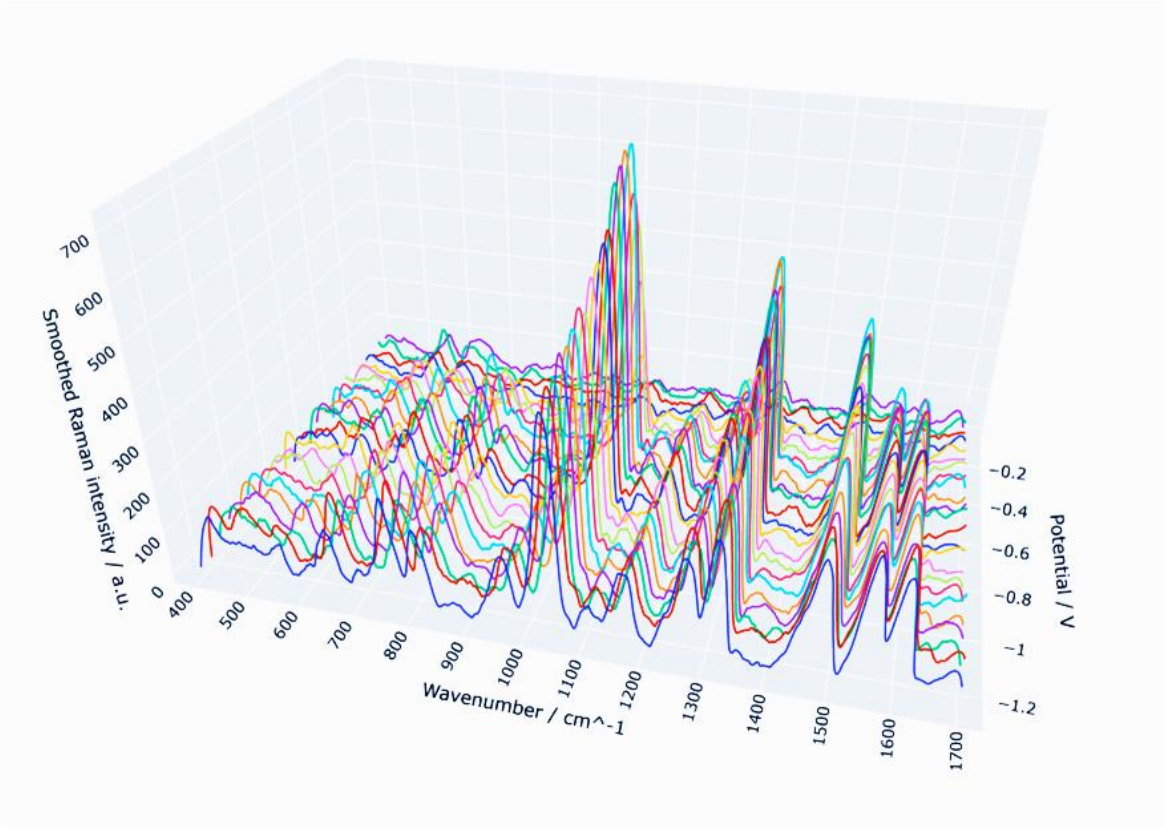


Figure 3: Interactive 3D waterfall plot of methylene blue Raman spectra recorded at different electrode potentials.

8.8 Expected outputs

- Processed Excel workbook
- Interactive 3D surface (HTML)
- Interactive heatmap (HTML)
- Interactive contour plot (HTML)
- Interactive rotatable waterfall plot (HTML)
- Optional raw-versus-smoothed comparison waterfall
- Static PNG/PDF plots, where enabled

8.9 Excel sheets

The exported workbook typically contains Metadata, Raw_Intensity_Matrix, smoothed_Intensity_Matrix, Plotted_Intensity_Matrix and the corresponding long-format tables. Current waterfall-enabled releases may also include shifted values, unshifted values and offset tables for custom plotting in Excel.

9. Absorpto- and fluorovoltammograms

9.1 Purpose

This combined module analyzes potential-resolved absorption or fluorescence matrices. The data structure is analogous to that of the Raman voltammogram module, but the spectral axis is wavelength rather than Raman shift, and the response values are absorbance or fluorescence intensity.

9.2 Select the analysis mode

Choose Absorption or Fluorescence so that titles, axis labels and output metadata match the selected measurement type. The underlying matrix orientation remains the same: potentials in the first row, wavelengths in the first column.

9.3 Wavelength range

Select only the spectral interval needed for interpretation. Restricting the range can reduce file size and improve visual resolution. Ensure that the selected interval includes the complete band shape when comparing peak positions or widths.

9.4 Waterfall data for Excel

In addition to the graphical waterfall output, the module exports numerical waterfall values for user-defined Excel plots. The shifted matrix contains the vertical offsets, the unshifted matrix preserves the processed spectral values, and a separate offset table documents the applied vertical offset.

9.5 Typical outputs

- Processed absorption or fluorescence matrix
- Interactive surface, heatmap and contour files
- Interactive and/or static waterfall plot

- Excel workbook containing raw/processed data, metadata and waterfall.

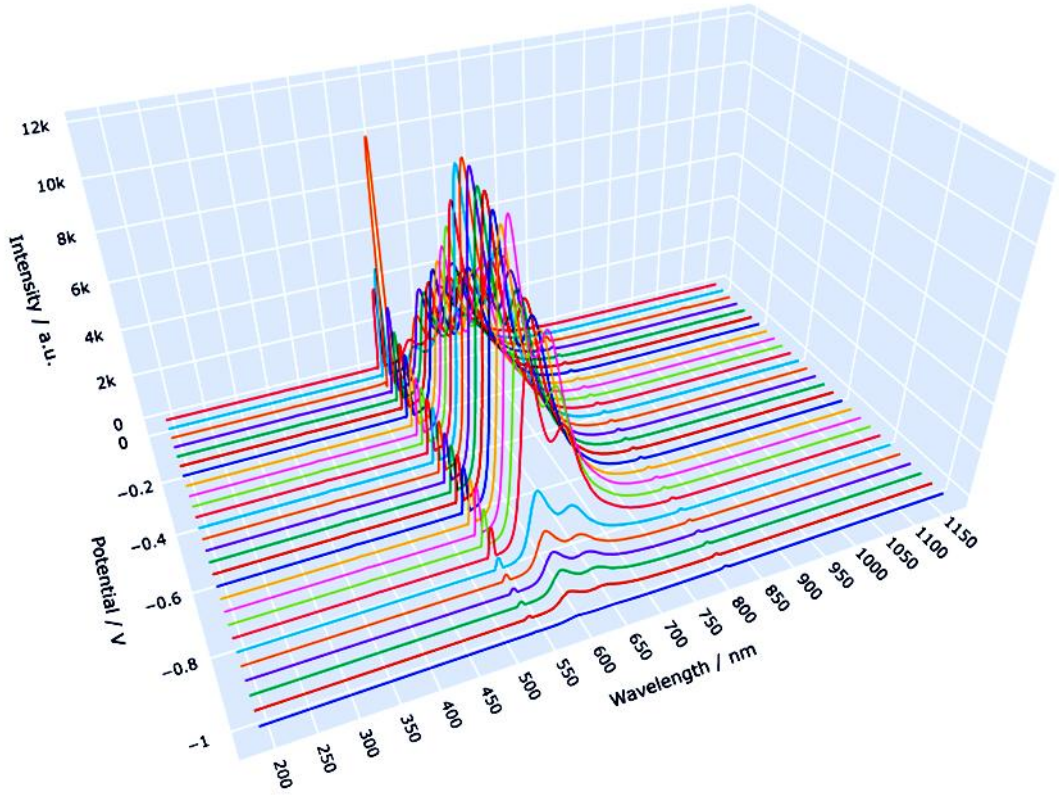


Figure 4: Fluorovoltammogram of rubrene at different potentials.

10. Rotating Ring–Disk Electrode (RRDE) Analysis

10.1 Introduction

The rotating ring–disk electrode (RRDE) is a well-established electrochemical technique for investigating electrochemical reaction mechanisms and soluble intermediates. In contrast to conventional Rotating Disk Electrode (RDE) experiments, the RRDE combines a central disk electrode with a concentric ring electrode. During an experiment, both electrodes rotate simultaneously at a precisely controlled speed.

Species generated at the disk electrode are transported by convection to the surrounding ring electrode, where they can be detected electrochemically. This enables simultaneous investigation of the primary electrochemical reaction occurring at the disk and the fate of soluble intermediates.

Typical applications include:

- Oxygen reduction reaction (ORR)
- Hydrogen peroxide detection
- Evaluation of catalyst selectivity
- Metal deposition and dissolution
- Electrochemical reaction mechanism studies

- Homogeneous redox mediators

Compared with conventional RDE measurements, RRDE experiments provide considerably more mechanistic information because both the main reaction and intermediate products can be monitored simultaneously.

10.2 Principle of the RRDE

The RRDE consists of two independently controlled working electrodes:

- **Disk electrode (WORKING ELECTRODE1):** where the primary electrochemical reaction occurs.
- **Ring electrode (WORKING ELECTRODE2):** detects products formed at the disk.

Both electrodes rotate with the same angular velocity. Rotation generates a well-defined hydrodynamic flow that transports dissolved species from the disk to the ring.

Only a fraction of the products generated at the disk reaches the ring. This fraction is called the **collection efficiency (N)**.

Typical collection efficiencies range from approximately 0.20 to 0.40, depending on the electrode geometry.

For the Metrohm Pt/GC RRDE (3.109.4080), the nominal collection efficiency is approximately **25 %**.

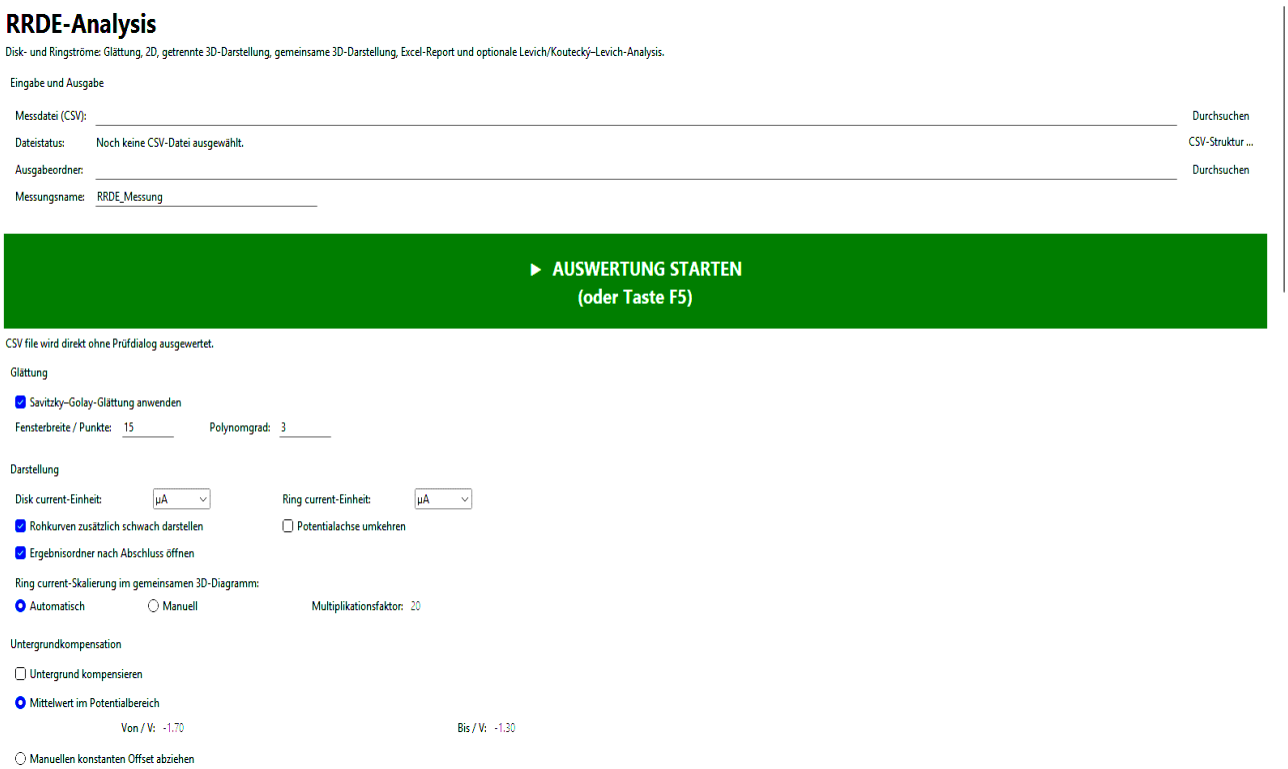


Figure 5: The first part of the input for the RRDE analysis.

10.3 Data Import

SpectroElectroChem Suite imports RRDE data directly from CSV files exported by NOVA.

The expected data structure is

Column	Content
1	Potential
2	Disk current (Rotation 1)
3	Ring current (Rotation 1)
4	Disk current (Rotation 2)
5	Ring current (Rotation 2)
...	Additional rotation speeds

The first row contains the rotation speed for each measurement pair.

Example:

Potential (V)	Disk 500 rpm	Ring 500 rpm	Disk 1000 rpm	Ring 1000 rpm
0.20	...	...	...	...

After import, the software automatically identifies

- rotation speed,
- disk current,
- ring current,
- measurement order.

No manual column assignment is required, which minimizes user interaction and reduces the risk of assignment errors.

10.4 Main RRDE Window

The RRDE module provides a graphical interface for importing, processing and evaluating RRDE measurements.

The main window contains the following sections:

Data Import

- Load CSV file
- Automatic recognition of rotation speeds
- Automatic assignment of disk and ring currents

Background Correction

The user may choose between

- no correction,
- manual background correction,
- potential-interpolated subtraction of a separate N_2/background RRDE measurement.

In version 6.0, background correction can be applied to both disk and ring currents. When a separate N_2/background RRDE measurement is used, the software matches the background data to the O_2 measurement by rotation rate and interpolates the background currents onto the potential grid of the corresponding O_2 measurement before subtraction.

Collection Efficiency

The collection efficiency **N** may either

- be entered manually,
- or taken from the electrode manufacturer's specifications.

Typical values range between 0.20 and 0.40.

10.5 Background Correction

In RRDE experiments, measured disk and ring currents may contain background contributions that are not part of the electrochemical process of interest. These contributions can influence quantitative evaluation, particularly when the ring current is small.

Examples include

- capacitive currents,
- oxidation of impurities,
- oxygen-independent reactions,
- electrode surface processes.

SpectroElectroChem Suite therefore provides two background-correction approaches.

Manual Background

A constant background current can be subtracted when the background is approximately independent of potential. This option is mainly intended for simple exploratory corrections.

For potential-dependent backgrounds, a separately measured N_2/background data set is preferable.

Nitrogen Background Measurement

A separate RRDE background measurement recorded under N_2 (or under otherwise appropriate background conditions) can be imported in addition to the O_2 measurement. The background file should contain the same rotation rates and corresponding disk/ring current pairs as the measurement to be corrected.

For each rotation rate, the software first checks the correspondence of the measurement series. Because the potential grids of the O_2 and N_2 measurements may differ slightly, the background currents are linearly interpolated onto the potential grid of the O_2 measurement before subtraction:

$I_{Disk,corr}(E) = I_{Disk,O_2}(E) - I_{Disk,N_2,interp}(E)$
$I_{Ring,corr}(E) = I_{Ring,O_2}(E) - I_{Ring,N_2,interp}(E)$

The corrected currents are then used for subsequent RRDE evaluation. The program validates the point count, scan direction, potential range and rotation-rate matching and reports incompatible data sets instead of silently subtracting mismatched curves.

Matching and interpolation

Background measurements are matched to the O_2 measurement by rotation rate. For every matched pair, the N_2/background disk and ring currents are interpolated as functions of potential onto the potential values of the O_2 curve. Subtraction is therefore performed at identical potentials even when the two measurements were recorded on slightly different potential grids.

Excel documentation

The RRDE Excel report documents the complete correction procedure in a dedicated background-correction worksheet. It includes the original O_2 currents, the interpolated N_2/background currents, the background-corrected disk and ring currents and, where smoothing is enabled, the smoothed-corrected currents. This preserves the numerical steps of the correction for independent inspection and reproducibility.
Scientific check: Background subtraction is only meaningful when O_2 and N_2/background measurements were recorded under comparable experimental conditions. In particular, electrode state, electrolyte composition, scan direction, rotation rates and potential range should be consistent.

10.6 Collection Efficiency

The collection efficiency N describes the fraction of reaction products formed at the disk that reaches the ring electrode.

It depends exclusively on the RRDE geometry.

Typical values are

- Metrohm Pt/Pt RRDE: approximately 0.25
- Metrohm Pt/GC RRDE: approximately 0.25

The user-defined collection efficiency is applied in all RRDE calculations. The ratio I_Ring/I_Disk may be used only under suitable calibration conditions to estimate an effective collection efficiency and should not be interpreted as its general definition. For quantitative RRDE analysis, use a collection efficiency determined from electrode geometry/manufacturer data or, preferably, verified experimentally with an appropriate reversible redox calibration system.

An incorrect value of N directly affects

- hydrogen peroxide yield,
- electron-transfer number,

- kinetic analysis.

Therefore the manufacturer-recommended value should normally be used.

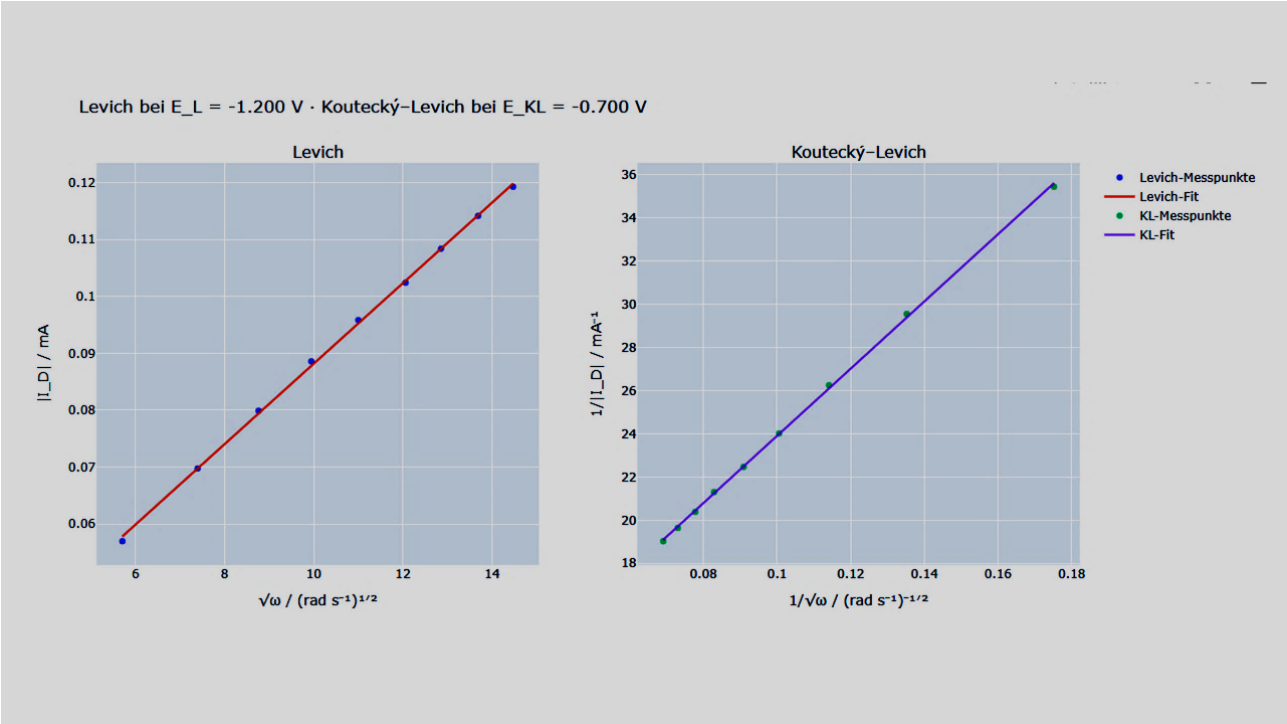


Figure 6: Levich (left) and Koutecký-Levich (right) graphs.

10.7 Electron-Transfer Number

One of the most important RRDE parameters is the apparent number of transferred electrons.

SpectroElectroChem Suite calculates

$n = 4 \cdot I_D / (I_D + I_R/N)$

where

- I_D = disk current
- I_R = ring current
- N = collection efficiency.

Typical interpretation:

n	Interpretation
≈4	Direct four-electron reduction
3–4	Mixed mechanism
≈2	Hydrogen peroxide formation dominates

For oxygen reduction, values close to four indicate efficient reduction directly to water (acidic media) or hydroxide ions (alkaline media), whereas values near two indicate predominant hydrogen peroxide formation.

10.8 Hydrogen Peroxide Yield

The percentage of hydrogen peroxide generated during oxygen reduction is calculated as

$$\%H_2O_2 = 200 \cdot (I_R/N) / (I_D + I_R/N)$$

The software calculates and displays the hydrogen peroxide yield as a function of potential and allows comparison of different catalyst materials.

Typical interpretation:

H_2O_2 yield	Interpretation
<5 %	Excellent four-electron catalyst
5–20 %	Mixed reaction pathway
>20 %	Significant peroxide production

10.9 Equilibrium Potential and Reference-Electrode Conversion

For kinetic evaluation, disk potentials are usually expressed as overpotentials relative to the thermodynamic equilibrium (reversible) potential of the reaction under study. SpectroElectroChem Suite can calculate this equilibrium potential automatically from the selected reaction, the pH and the temperature, and convert it to the reference electrode actually used in the experiment.

The pH dependence follows the Nernstian slope $2.303 \cdot R \cdot T/F$ per pH unit (approximately 59 mV per pH unit at 25 °C). The user selects the reaction (for example oxygen reduction, oxygen evolution, hydrogen evolution or hydrogen oxidation), the reference electrode, the pH and the temperature; alternatively a user-defined equilibrium potential can be entered directly.

Scientific check: Verify that the chosen reaction, pH and reference electrode correspond to the real experimental conditions. An incorrect equilibrium potential shifts the entire overpotential axis and therefore the apparent exchange current and Tafel intercept, even though the Tafel slope itself is unaffected.

10.10 Levich and Koutecký–Levich Analysis

The dependence of the diffusion-limited disk current on rotation rate is evaluated with the Levich and Koutecký–Levich relations. In the Levich representation, the limiting current is plotted against the square root of the angular velocity, $\sqrt{\omega}$. Under the assumptions of the Levich model, linearity is expected for diffusion-controlled transport. In the Koutecký–Levich representation, reciprocal current is plotted against $1/\sqrt{\omega}$; the intercept is used to estimate the kinetic current.

The software determines the limiting current from a user-selected diffusion-limited plateau and reports the plateau drift, the relative scatter and the plateau slope as quality indicators. Absolute cathodic disk currents are used so that reduction currents are handled consistently. Always inspect the selected plateau region before accepting the analysis: a sloping or drifting plateau indicates that a true limiting current has not been reached.

10.11 Tafel Analysis, Exchange Current and Butler–Volmer Parameters

The Tafel module fits the linear region of the potential–log|current| relationship for each rotation rate and reports the Tafel slope, the intercept and the coefficient of determination R^2. From the intercept and the equilibrium potential the exchange current, or the exchange current density, is derived.

Current density option. Currents may be evaluated either as measured currents or as current densities. When the current density mode is selected, the disk currents are divided by the geometric electrode area (default 0.196 cm^2, corresponding to a 5 mm diameter disk). Enter the area that matches the electrode actually used.

Mass-transport correction. Optionally, the Tafel region can be corrected for mass transport using the Koutecký–Levich relation, so that the extracted kinetic current more closely reflects the charge-transfer kinetics rather than the combined kinetic and diffusion response. This requires a valid limiting-current plateau (see Section 10.10).

Standard heterogeneous rate constant. If requested, the standard rate constant k_0 is estimated from the exchange current density through the Butler–Volmer relation

$j_0 = n \cdot F \cdot k_0 \cdot c_O^{(1-\alpha)} \cdot c_R^{\alpha}$, using the entered electron number n, transfer coefficient α, and bulk concentrations of the oxidized and reduced species. This calculation is only meaningful for a simple, well-defined one-step redox couple; it is not applied to multi-electron reactions such as ORR, OER, HER or HOR, for which the Butler–Volmer concentration terms are not defined in this simple form.

The Tafel Excel report contains a structured summary sheet with the general settings, the per-curve Tafel results, the limiting-current table (rotation rate, limiting current, drift, relative scatter and plateau slope) and a dedicated section collecting all quality warnings.

10.12 Graphical Output

SpectroElectroChem Suite automatically generates:

- Disk current vs. potential
- Ring current vs. potential
- Combined disk/ring plots
- Hydrogen peroxide yield
- Electron-transfer number
- Interactive three-dimensional RRDE plots
- Excel export of original, background, corrected and processed data

Static figures can be exported at high resolution for further preparation of publication figures.

Interactive HTML figures allow free rotation, zooming and panning.

By convention, disk currents are reported in milliamperes (mA) and ring currents in microamperes (µA), reflecting the much smaller magnitude of the ring current. This convention is applied consistently in the plots, the combined data sheet and the Excel charts.

Static PNG and PDF graphics are intended for publication-quality figures.

10.13 Best Practice

For reliable RRDE measurements the following recommendations are suggested:

- Polish the disk electrode before each experiment.
- Ensure that the ring electrode is clean.
- Use freshly prepared electrolyte.
- Remove dissolved oxygen when recording background measurements.
- Use identical rotation speeds for all comparative measurements.
- Verify the collection efficiency before quantitative analysis.
- Inspect the limiting current plateau before performing Levich or Koutecký–Levich analysis.

10.14 Worked example: oxygen reduction (ORR) in alkaline solution

This example illustrates a complete RRDE workflow for the oxygen reduction reaction and the values a user would typically enter.

Experimental setup. Glassy-carbon disk carrying the catalyst film and a platinum ring; collection efficiency N = 0.25. Electrolyte: 0.1 M KOH saturated with O_2. The ring is held at a constant potential at which peroxide (HO_2^-) is re-oxidised, for example +0.40 V vs. RHE. The disk is swept from +1.0 V to +0.1 V vs. RHE at 10 mV s^{-1}. Rotation rates: 400, 900, 1600 and 2500 rpm.

Procedure in the software. (1) Import the NOVA CSV file. (2) Subtract an N_2 background for the ring current. (3) Enter the collection efficiency N = 0.25. (4) Select the potential window used for the peroxide evaluation. (5) Run the Levich and Koutecký–Levich analysis over the diffusion-limited plateau. (6) Optionally run the Tafel analysis using the ORR equilibrium potential for the chosen pH.

Typical results and interpretation. For a well-behaved four-electron catalyst the disk limiting current scales linearly with $\sqrt{\omega}$ in the Levich plot, the electron number n(E) computed from the disk and ring currents lies close to 4 across the plateau, and the hydrogen-peroxide yield stays below about 5 %. Values of n approaching 2 with a rising $\%H_2O_2$ indicate a two-electron pathway. The Koutecký–Levich intercept gives the kinetic current, and the Tafel slope characterises the charge-transfer kinetics.

11. Spectro-CV Synchronization

11.1 Purpose

The Spectro-CV Synchronization module combines cyclic-voltammetry (CV) data with independently recorded time-resolved spectra. It is designed for absorption, fluorescence/ECL (wavelength-resolved) and Raman experiments in which the potentiostat and spectrometer acquire data at different sampling rates. The central task is to place each spectrum on the electrochemical time/potential axis without altering the original electrochemical data.

11.2 Input data

Two CSV files are loaded: one electrochemical file and one spectral file. The CV file contains potential in the first column and one or more current columns. The spectral file contains the spectral coordinate (wavelength or Raman shift) and the intensity matrix. The module accepts the original, non-transposed spectral arrangement in which the spectral coordinate is stored in the first column; internally the matrix is oriented for synchronization and plotting.
For reproducible synchronization, enter the experimental CV information used during acquisition, in particular the potential limits, scan direction, number of cycles and, where required by the selected import path, scan rate or potential increment. The number of recorded spectra is allowed to differ from the number of electrochemical points.

11.3 Synchronization principle

The electrochemical measurement defines a time-ordered sequence E(t) and I(t). Each recorded spectrum has its own acquisition time, determined by the spectrometer timing and the acquisition order. The program maps the spectral acquisition times onto the CV time axis and assigns the corresponding potential by interpolation. This preserves turning points and repeated cycles provided that the experimental sequence and cycle count are entered correctly.
When the spectra are acquired at constant temporal spacing, the potential associated with spectrum k is obtained from the interpolated CV trajectory at the corresponding acquisition time. The method therefore does not require the number of spectra to equal the number of CV points.

11.4 Absorption and fluorescence workflows

For absorption and fluorescence data, the user selects the wavelength interval used for visualization and export. Individual wavelengths can be selected interactively to generate potential-dependent optical traces. Several wavelengths may be selected simultaneously, allowing direct comparison of growing and decaying bands during a redox cycle.

- Preview the synchronized CV and optical response before export.
- Select one or more wavelengths from the spectrum display.
- Use smoothing only when required by the signal-to-noise ratio and verify that band maxima and turning-point behaviour are not displaced.
- Export all cycles when cyclic reproducibility is of interest; mean-cycle representations should be treated as derived data rather than replacements for the individual cycles.

11.5 Raman workflow and baseline correction

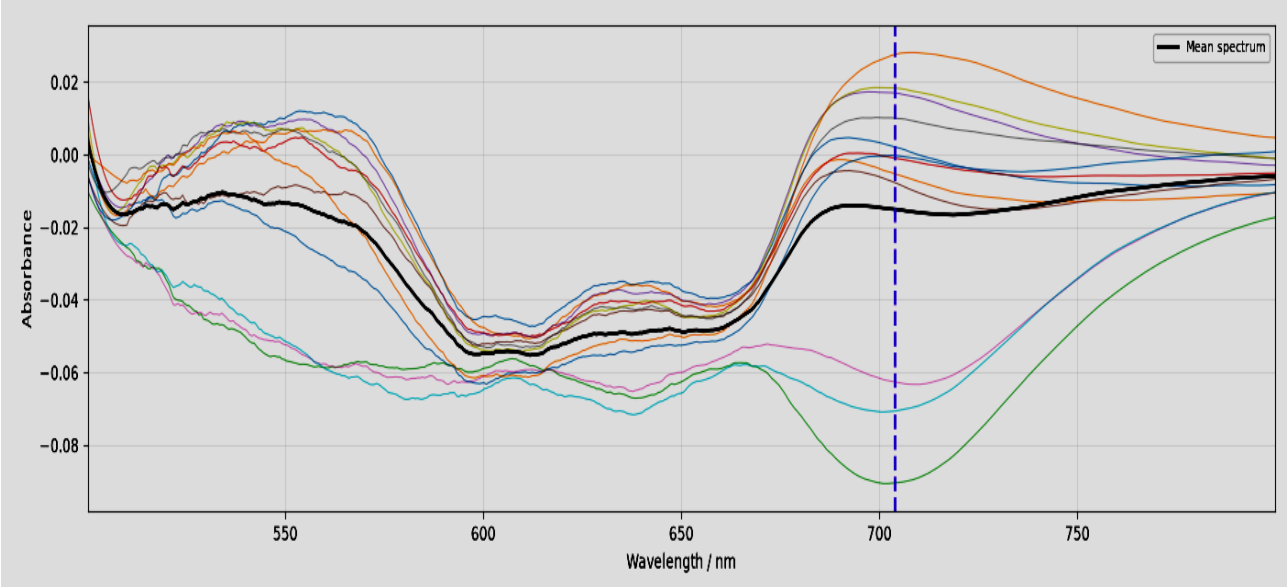
Raman data can be synchronized in the same way, using Raman shift as the spectral coordinate. Because Raman spectra frequently contain a broad fluorescence background, the module provides baseline correction before peak-oriented evaluation. Depending on the selected option, penalized baseline methods such as AsLS or arPLS can be used. The baseline parameters should be chosen so that the slowly varying background is removed without suppressing broad Raman bands.
Peak selection is interactive: peaks can be detected, individual peaks removed from the selection and spectral regions restricted to the range relevant for interpretation. Always inspect several representative spectra from different potentials after baseline correction.

11.6 Potential-dependent spectral derivatives

For selected wavelength or Raman-shift traces, the module can calculate the finite-difference derivative of the optical response with respect to potential. For absorbance A, the local derivative is evaluated as

$$dA/dE \approx (A_{i+1} - A_i) / (E_{i+1} - E_i)$$

The derivative emphasizes potential regions in which the optical signal changes rapidly. Its sign depends on scan direction and on whether the selected band increases or decreases. Near CV reversal potentials, finite-difference derivatives require particular care because the direction of E changes. Derivative curves should therefore always be interpreted together with the original synchronized trace.



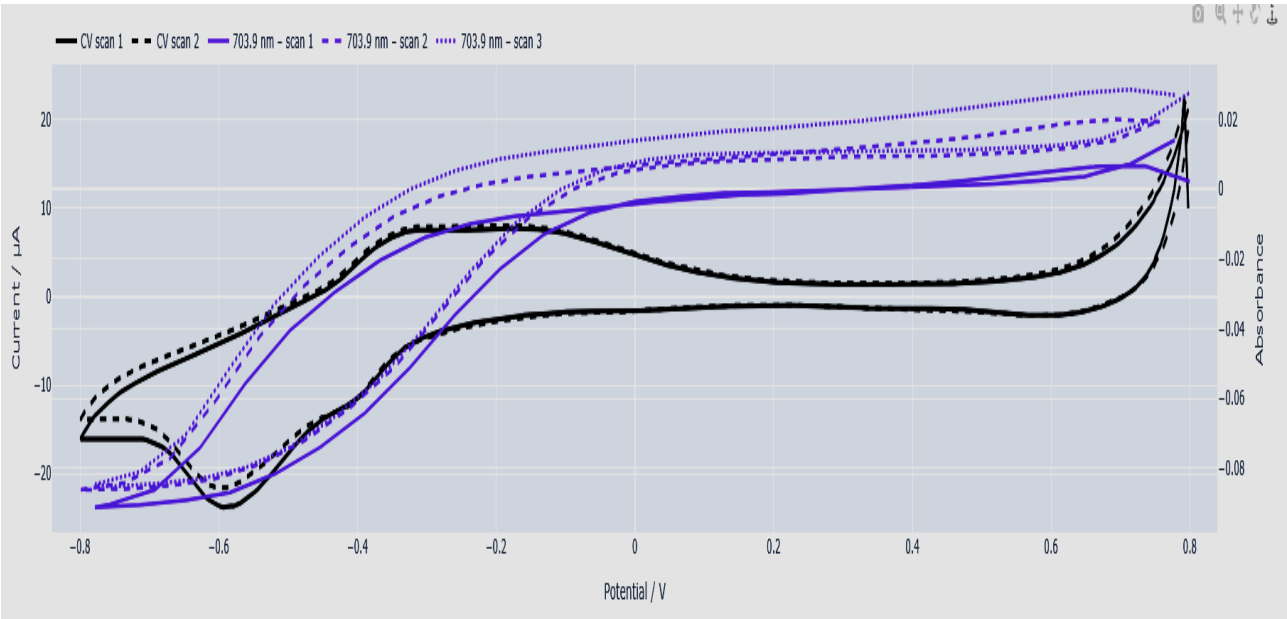


Figure 7: Top: Change in methylene blue absorbance as a function of applied potential. Bottom: Absorptovoltammogram for the selected wavelength (703.9 nm).

11.7 Multiple cycles and mean curves

For experiments containing several CV cycles, the synchronized data can be retained cycle by cycle. Where implemented for the selected analysis type, mean CV and mean spectral responses can additionally be calculated from corresponding points of the individual cycles. A mean curve is useful for visualization of reproducible behaviour, but cycle-to-cycle differences should be inspected before averaging.

11.8 Graphical and numerical outputs

- Synchronized CV with selected optical or Raman traces, using a secondary ordinate where appropriate.
- Static PNG figures for documentation and publication workflows.
- Interactive 3D waterfall plots of the synchronized spectra.
- Excel workbooks containing synchronized potentials, currents, spectral values, selected traces and derived quantities.
- For Raman analysis, baseline-corrected spectra, selected peak information and mean-cycle data where enabled.

11.9 Scientific checks

Scientific check: Synchronization cannot correct an incorrect experimental sequence. Verify the number of CV cycles, scan direction, start and reversal potentials, spectrometer acquisition order and total acquisition time. A visually smooth synchronized plot is not proof of correct timing. Whenever possible, confirm synchronization using a spectral event that is expected close to a known electrochemical feature.

12. ECL Synchronization (Integrated Signal)

12.1 Purpose

This module synchronizes cyclic voltammetry with a non-wavelength-resolved electrogenerated chemiluminescence (ECL) signal, for example a photomultiplier or photodiode output. In contrast to wavelength-resolved fluorescence/ECL in Chapter 11, the optical channel is treated as a single intensity-versus-time signal. The module supports repeated CV cycles, cycle means, numerical integration and concentration-series evaluation.

12.2 Input structure

The electrochemical and ECL signals may be stored in the same CSV file as separate blocks or imported through the file layout supported by the current module. The electrochemical block contains potential and current traces; the ECL block contains its own potential or time-related axis and the ECL traces. The sampling grids do not need to be identical because synchronization is performed numerically.
For repeated measurements, enter the number of cycles. For concentration-series experiments, also enter the number of concentrations and the corresponding values in mol/L. Scientific notation, including values such as 1e-6 mol/L, can be used.

12.3 Synchronization and cycle means

The program aligns the electrochemical and ECL acquisition sequences to a common electrochemical trajectory. Individual cycles remain available for inspection. Mean CV and mean ECL curves are then calculated from corresponding cycle positions so that both the repeatability and the averaged response can be evaluated.

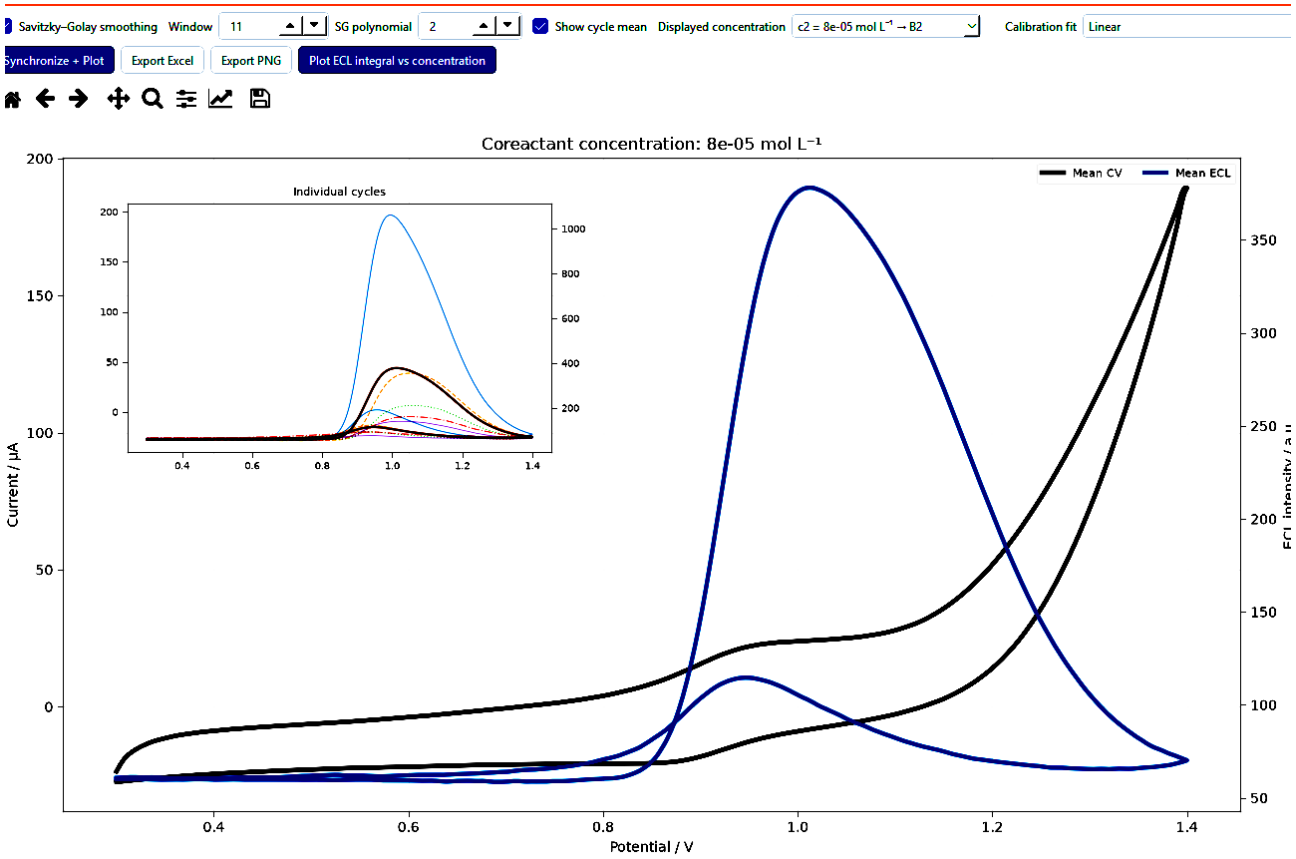


Figure 8: CV (black) and ECL (blue) curves. Average of five scans. Insert: Five individual scans.

12.4 Integrated ECL intensity

For each ECL cycle, the integrated signal is calculated numerically over the synchronized acquisition interval:

$$S_ECL = \int I_ECL(t)\, dt$$

The result has units of signal × time and represents the total detected light response for the selected cycle. When several cycles are present, the module reports the individual integrals and their mean. Integration is particularly useful when the temporal ECL profile changes shape while the total emitted signal remains the quantity of interest.

12.5 Comparison with and without co-reactant

The module can compare measurements recorded without and with a co-reactant. Because absolute ECL intensity depends on detector sensitivity, optical geometry and acquisition settings, both data sets should be recorded under identical instrumental conditions. Ratios of integrated signals are meaningful only when the same integration interval and scaling are used.

12.6 Concentration series and calibration

For a series of known concentrations, the mean integrated ECL intensity is plotted as a function of concentration. The user can choose a linear fit or a polynomial fit of the degree available in the interface. The regression equation and coefficient of determination R^2 are reported and exported. A polynomial model should not be selected merely because it yields a larger R^2. Use the lowest-order model consistent with the known response mechanism and the residual structure, and avoid extrapolation beyond the calibrated concentration range.

12.7 Determination of an unknown concentration

An additional ECL data set with unknown concentration can be integrated using the same procedure. Its mean integrated intensity is evaluated against the selected calibration function. The resulting concentration is displayed graphically on the calibration plot and included in the export. If the calibration is non-monotonic over the fitted range, mathematical inversion may not be unique; restrict the calibration interval to a physically meaningful monotonic region.

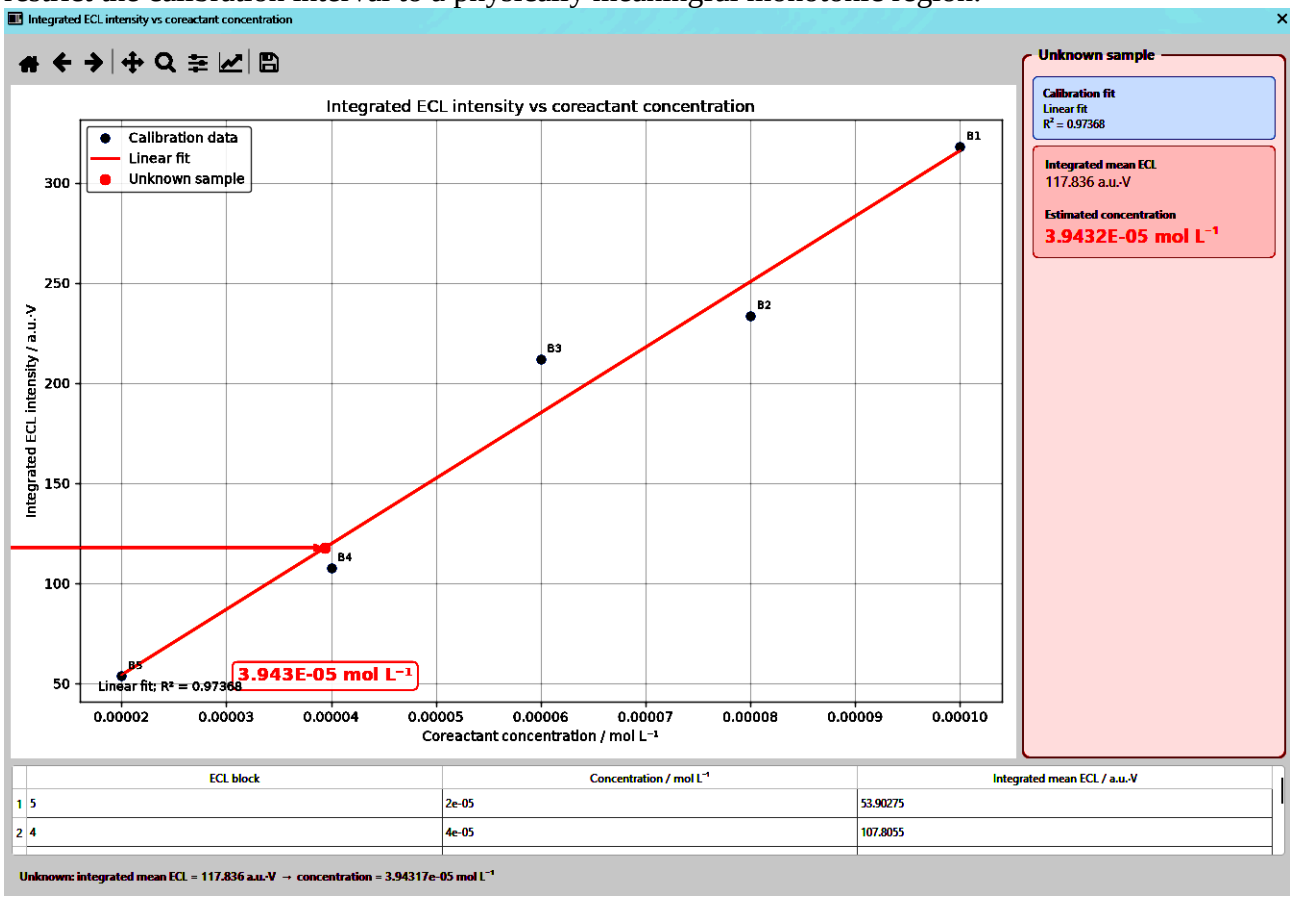


Figure 9: Integral ECL intensities as a function of concentration. Red arrow: Calculating an unknown concentration via ECL.

12.8 Outputs

- Preview of synchronized CV and ECL traces, including individual cycles and mean curves.
- Integrated ECL values for every cycle and mean integral values.
- Calibration plot: integrated ECL intensity versus concentration with equation and R^2.
- Graphical indication and numerical result for an unknown concentration, where used.
- PNG figures and Excel workbooks containing numerical data, integrals, fit coefficients and fit quality.

12.9 Scientific checks

Scientific check: ECL intensity is sensitive to electrode history, oxygen content, detector gain, cell geometry and co-reactant concentration. Before interpreting a calibration, inspect cycle-to-cycle repeatability, confirm identical acquisition settings and verify that the integrated signal is not dominated by baseline drift or detector saturation.

13. Diffusion Coefficient Analysis

13.1 Purpose

The Diffusion Coefficient Analysis module determines diffusion coefficients D from three classical electrochemical approaches: Cottrell analysis of chronoamperometry, Anson analysis of chronocoulometry and Randles–Ševčík analysis of cyclic voltammetry recorded at different scan rates. The three methods provide complementary estimates and have different sensitivities to charging current, adsorption, electrode kinetics and peak selection.

13.2 Required parameters and units

Enter the concentration c, number of transferred electrons n, geometric electrode area A and temperature T required by the selected method. The output diffusion coefficient is reported in cm^2/s when the electrochemical quantities are supplied in a consistent centimetre-based unit system. Concentrations entered in mol/L must be converted internally to mol/cm^3 where the classical equations require it; 1 mol/L = 10^{-3} mol/cm^3.

13.3 Cottrell analysis

For a planar electrode under diffusion control after a potential step, the Cottrell equation is

$$I(t) = n\,F\,A\,c\,\sqrt{D} / \sqrt{(\pi\,t)}$$

Accordingly, a plot of current I versus $t^{-1/2}$ should be linear in the diffusion-controlled interval. If the fitted slope is m_C,

$$D = \pi\,[m_C / (n\,F\,A\,c)]^2$$

The fitting interval is selected by the user. Very early times are often affected by double-layer charging, uncompensated resistance and the finite response time of the potentiostat, whereas very long times may deviate from semi-infinite planar diffusion. Select the range on the basis of linearity and physical plausibility, not only R^2.

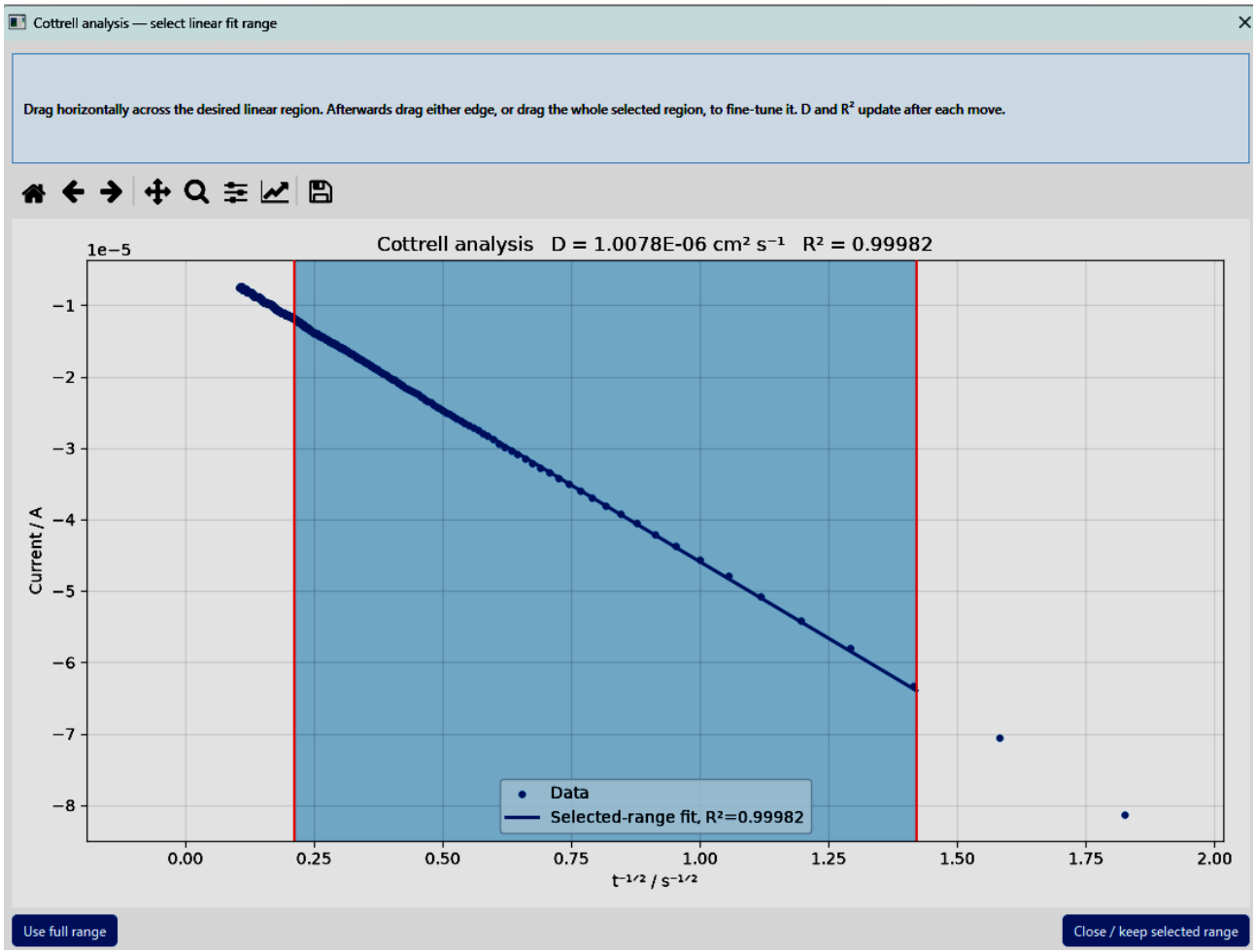


Figure 10: Cottrell analysis. The current I is plotted against $t^{-1/2}$, and the linear fit interval is selected interactively (shaded band). The diffusion coefficient is obtained from the slope (here $D = 1.01 \times 10^{-6}$ cm² s⁻¹, $R^2 = 0.9998$). Points affected by double-layer charging at very short times fall outside the selected range.

13.4 Anson analysis

Chronocoulometry integrates the current and can reduce the visual influence of high-frequency current noise. For diffusion-controlled charge transfer at a planar electrode, the Anson relation is

$$Q(t) = 2\,n\,F\,A\,c\,\sqrt{(D\,t/\pi)} + Q_{dl} + n\,F\,A\,\Gamma$$

where Q_dl is the double-layer charging contribution and Γ represents an adsorbed amount when present. A plot of Q versus $t^{1/2}$ is linear for ideal diffusion control. For slope m_A,

$$D = \pi\,[m_A / (2\,n\,F\,A\,c)]^2$$

The intercept is not part of the diffusion coefficient calculation but may contain physically relevant charging and adsorption contributions.

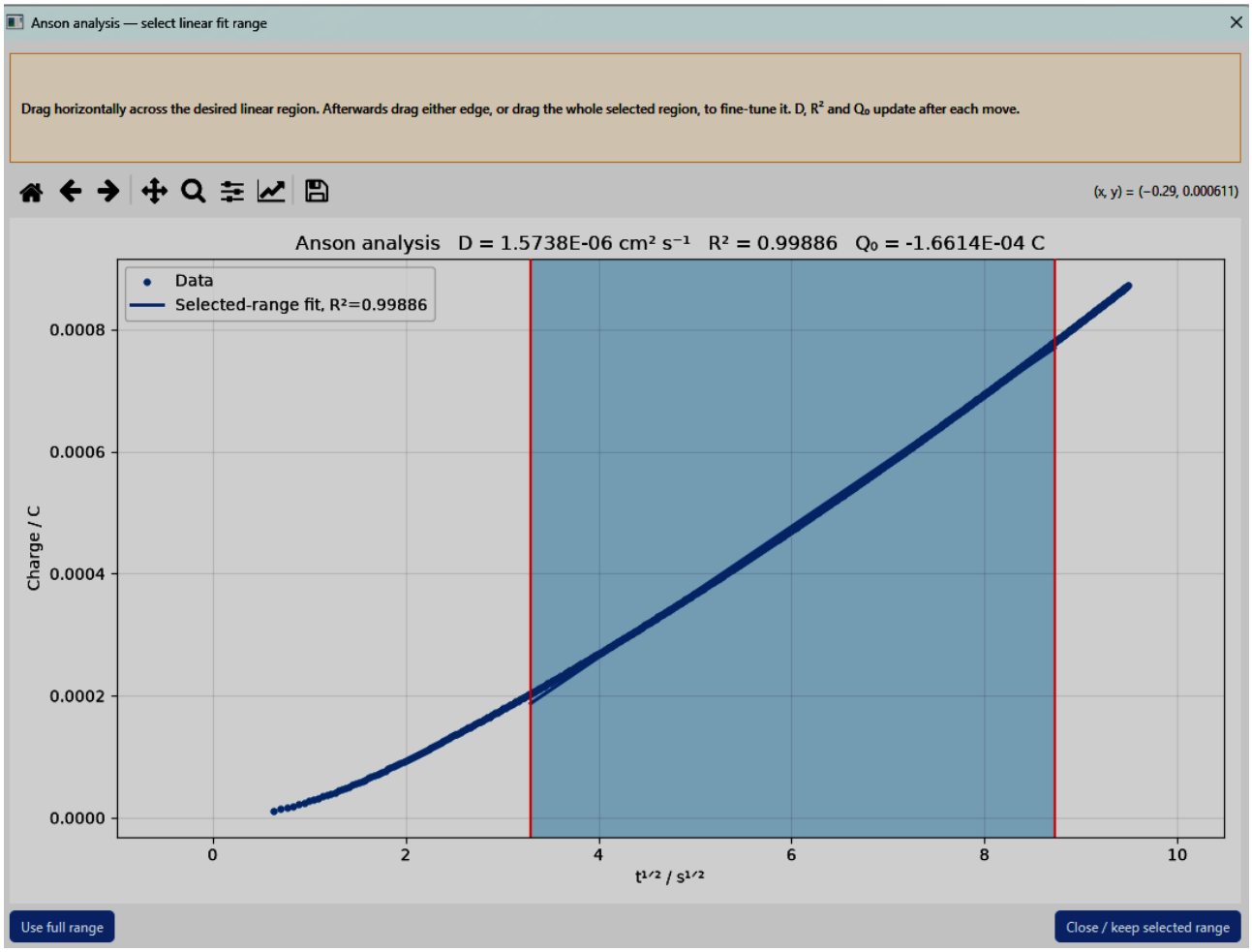


Figure 11: Anson analysis. The charge Q is plotted against $t^{1/2}$ with the linear fit interval selected interactively (shaded band). The diffusion coefficient follows from the slope (here $D = 1.57 \times 10^{-6}$ $cm^2\ s^{-1}$, $R^2 = 0.9989$); the intercept Q_0 collects the double-layer charging and any adsorption contribution.

13.5 Randles–Ševčík analysis

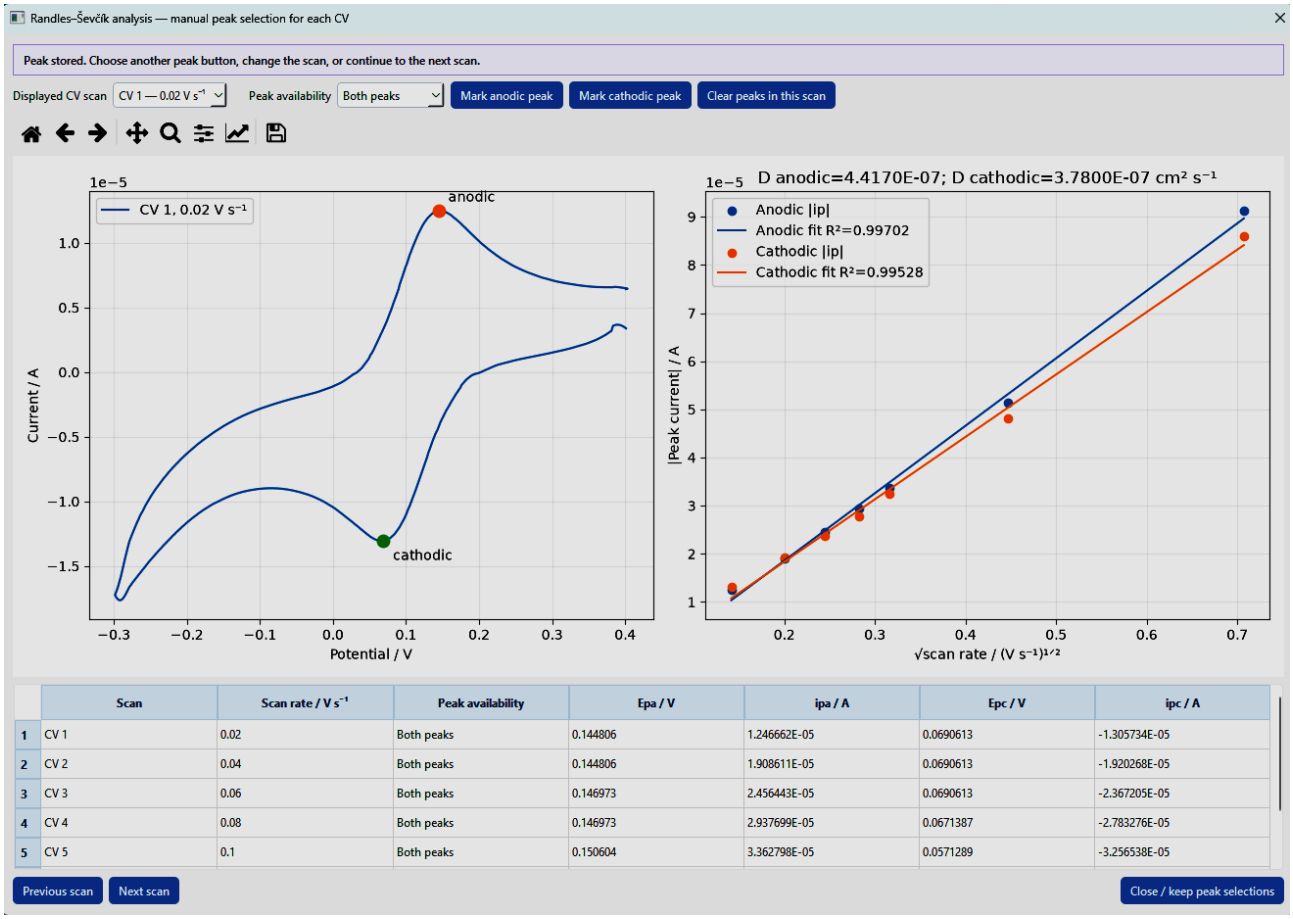
For a reversible, diffusion-controlled redox process at a planar electrode, the peak current depends on the square root of scan rate. The general Randles–Ševčík expression is

$$i_p = 0.4463\ n\ F\ A\ c\ \sqrt{(n\ F\ D\ v\ /\ R\ T)}$$

At 298.15 K, with A in cm², D in cm²/s, c in mol/cm³ and v in V/s, this becomes approximately

$$i_p = 2.69 \times 10^5\ n^{3/2}\ A\ c\ D^{1/2}\ v^{1/2}$$

The module extracts anodic or cathodic peak currents from CVs recorded at different scan rates and fits i_p versus $v^{1/2}$. The user can choose the peak direction relevant to the experiment. Inspect every detected peak before accepting the regression, especially when baseline currents or multiple redox peaks are present.



	Scan	Scan rate / V s⁻¹	Peak availability	Epa / V	ipa / A	Epc / V	ipc / A
1	CV 1	0.02	Both peaks	0.144806	1.246662E-05	0.0690613	-1.305734E-05
2	CV 2	0.04	Both peaks	0.144806	1.908611E-05	0.0690613	-1.920268E-05
3	CV 3	0.06	Both peaks	0.146973	2.456443E-05	0.0690613	-2.367205E-05
4	CV 4	0.08	Both peaks	0.146973	2.937699E-05	0.0671387	-2.783276E-05
5	CV 5	0.1	Both peaks	0.150604	3.362798E-05	0.0571289	-3.256538E-05

Figure 12: Randles–Ševčík analysis with manual peak selection. Left: a single cyclic voltammogram with the marked anodic (red) and cathodic (green) peaks. Right: the absolute peak current $|i_p|$ versus the square root of the scan rate for both branches, with linear regressions yielding separate anodic and cathodic diffusion coefficients (here D(anodic) = 4.42 × 10⁻⁷ cm² s⁻¹ and D(cathodic) = 3.78 × 10⁻⁷ cm² s⁻¹). The table lists the peak potentials and currents for each scan.

13.6 Selection of fitting ranges and peaks

Interactive interval selection is used for the Cottrell and Anson fits. For Randles–Ševčík analysis, the individual cyclic voltammograms should be inspected rather than relying only on an averaged CV. An apparently linear regression may still be physically invalid if the peak potential shifts strongly with scan rate, if the peak is kinetically limited or if uncompensated resistance dominates.

13.7 Comparison of the three methods

Agreement among Cottrell, Anson and Randles–Ševčík estimates supports, but does not prove, a diffusion-controlled model. Systematic differences can arise from electrode adsorption, coupled chemical reactions, non-reversible electron transfer, convection, finite diffusion geometry or inaccurate electrode area/concentration values. Report the method and fitting interval together with D.

13.8 Outputs

- Cottrell plot I versus $t^{-1/2}$ with selected fit interval, slope and D.
- Anson plot Q versus $t^{1/2}$ with selected fit interval, slope and D.
- Randles–Ševčík plot i_p versus $v^{1/2}$ for anodic or cathodic peaks with regression and D.
- Numerical results and fit data exported to Excel, together with the parameters used for the calculation.
- Static plots for reporting and comparison of the diffusion-coefficient estimates.

13.9 Nicholson Analysis

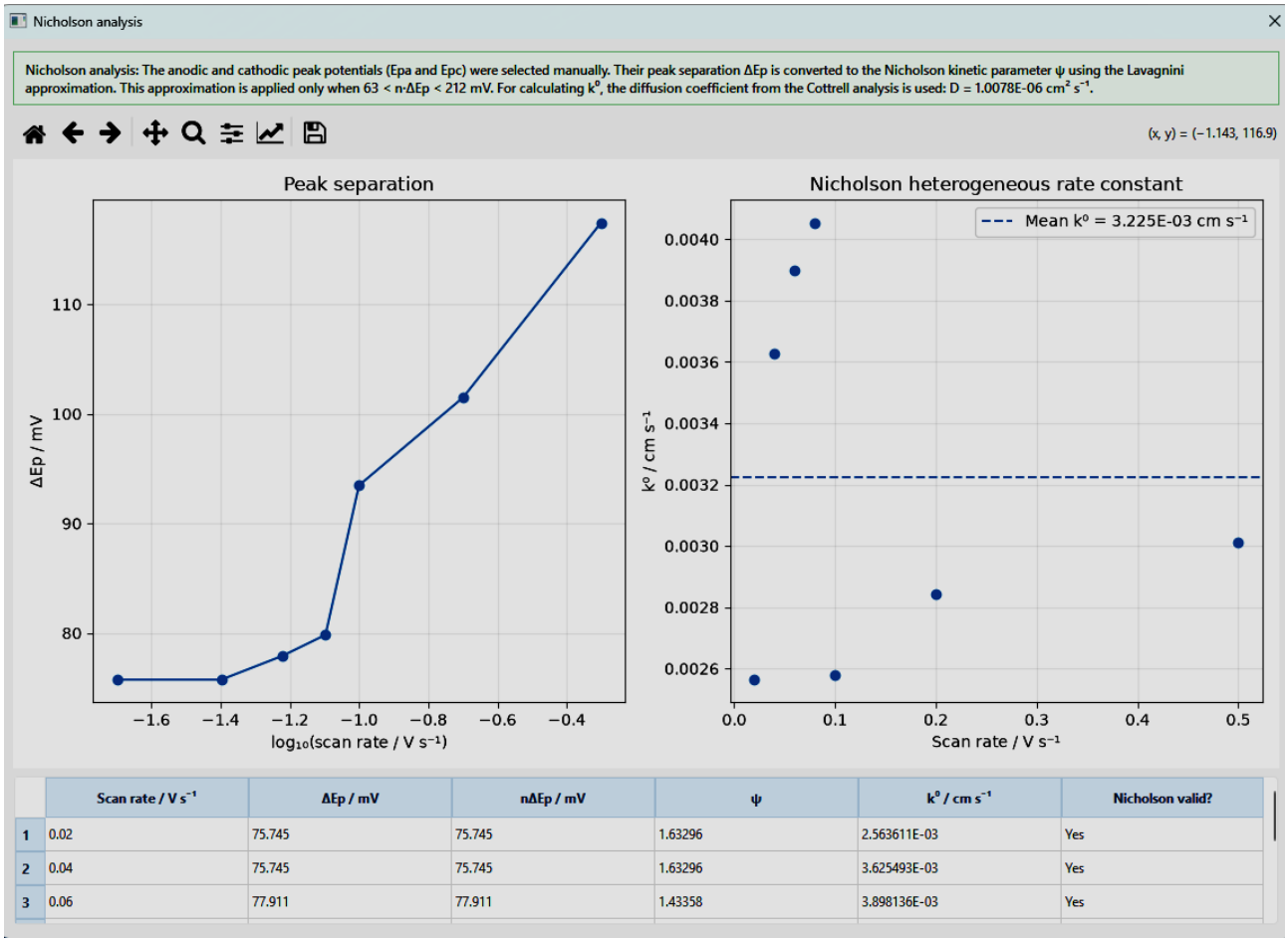


Figure 13: Nicholson analysis of the electron-transfer kinetics. Left: the anodic–cathodic peak separation ΔE_p as a function of $\log_{10}$(scan rate). Right: the heterogeneous rate constant k^0 derived from ΔE_p, together with its mean value (here $k^0 = 3.23 \times 10^{-3}$ cm s^{-1}). The diffusion coefficient from the Cottrell analysis is used as input, and the approximation is applied only where 63 mV < $n\Delta E_p$ < 212 mV.

13.10 Scientific checks

Scientific check: A diffusion coefficient is meaningful only if the assumptions of the selected model are approximately satisfied. Check electrode area, concentration units, electron number, temperature, fitting interval, scan-rate dependence and the presence of convection or adsorption. Do

not average discrepant D values from different methods without first identifying the origin of the discrepancy.

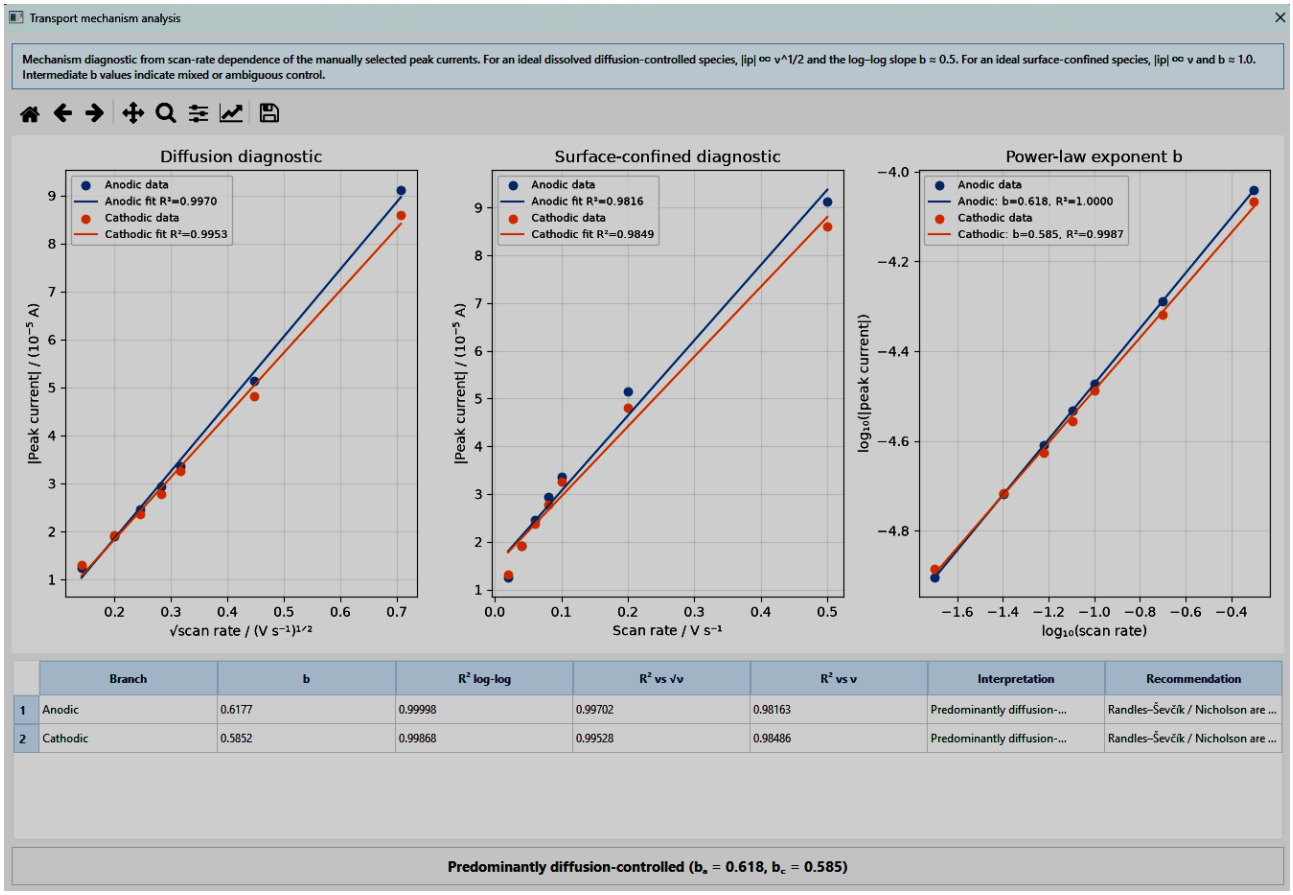


Figure 14: Transport-mechanism diagnostic based on the scan-rate dependence of the selected peak currents. Left: diffusion diagnostic, $|i_p|$ versus $\sqrt{v}$. Centre: surface-confined diagnostic, $|i_p|$ versus v. Right: log–log plot giving the power-law exponent b. A value of b ≈ 0.5 indicates an ideal dissolved, diffusion-controlled species and b ≈ 1.0 a surface-confined species; here b(anodic) = 0.62 and b(cathodic) = 0.59, indicating predominantly diffusion-controlled behaviour.

14. EIS Analysis

14.1 Purpose

The EIS Analysis module evaluates electrochemical impedance spectroscopy (EIS) data using Nyquist and Bode representations and fits a Randles-type equivalent circuit where appropriate. The module is intended to provide both a visual description of the frequency-dependent impedance and a quantitatively inspectable model fit, including residuals and Excel export.

14.2 Impedance data

EIS measurements are frequency dependent and are represented by the complex impedance

$$Z(\omega) = Z'(\omega) + j\,Z''(\omega)$$

where Z' is the real component, Z'' the imaginary component and $j^2 = -1$. The input file should contain frequency and the corresponding impedance components in the arrangement recognized by the module. Preserve the original frequency order and units from the instrument export.

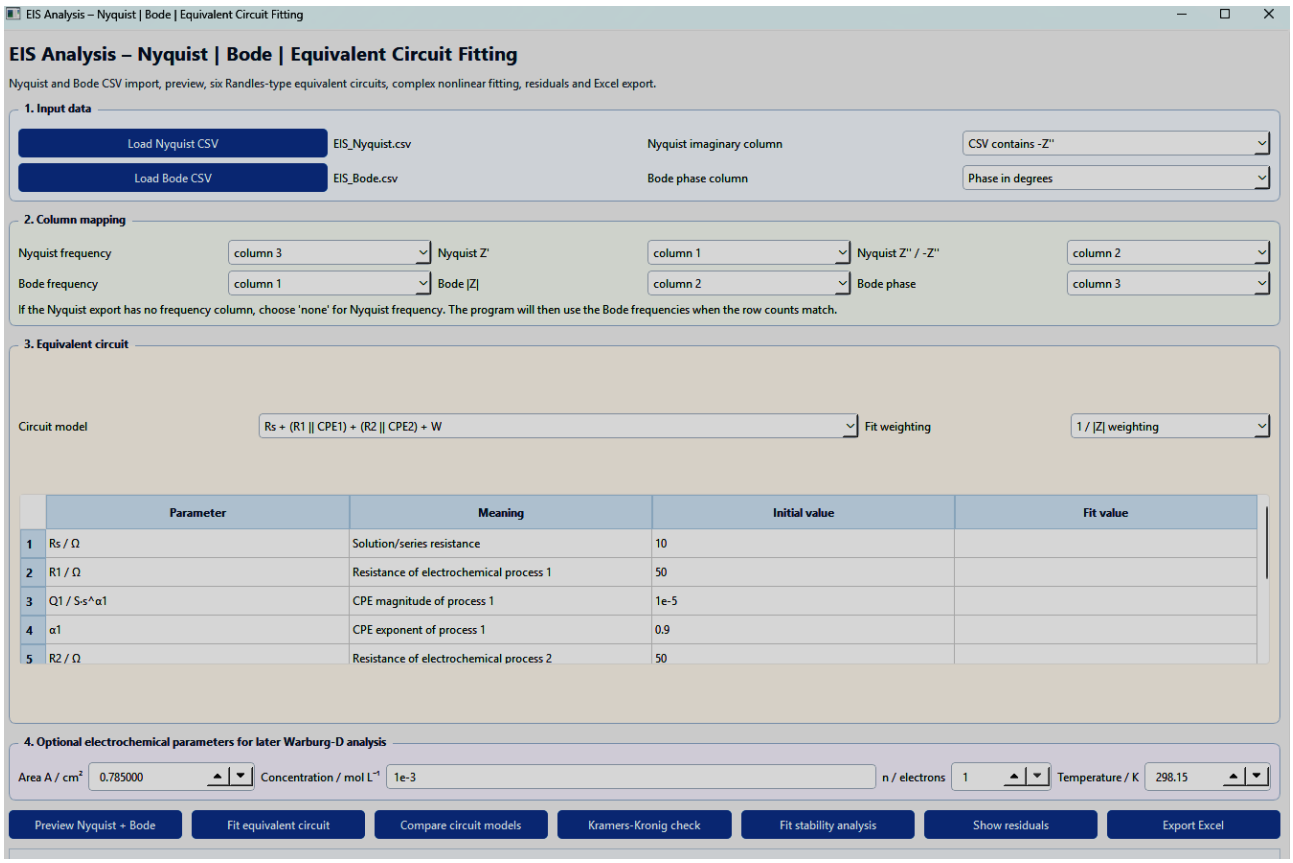


Figure 15: Main window of the EIS Analysis module. It provides CSV import for Nyquist and Bode data, column mapping, selection of one of the Randles-type equivalent circuits (here Rs + (R1 ∥ CPE1) + (R2 ∥ CPE2) + W), the table of initial parameters, and optional electrochemical parameters for a subsequent Warburg-based diffusion analysis.

14.3 Nyquist plot

The Nyquist representation plots the real impedance Z' on the abscissa and conventionally $-Z''$ on the ordinate. Frequency is implicit rather than shown as an axis. A semicircular feature is often associated with a charge-transfer process, whereas a low-frequency diffusion contribution can produce an inclined tail. These visual associations are qualitative until supported by a physically justified model.

14.4 Bode plots

The Bode representation displays impedance magnitude and phase angle as functions of frequency, usually on a logarithmic frequency axis:

$$|Z| = \sqrt{(Z'^2 + Z''^2)}$$
$$\varphi = \mathrm{atan2}(Z'', Z')$$

Bode plots are especially useful for identifying distinct time constants that may overlap in the Nyquist plane. Inspect both magnitude and phase; fitting only one representation can conceal systematic deviations in the other.

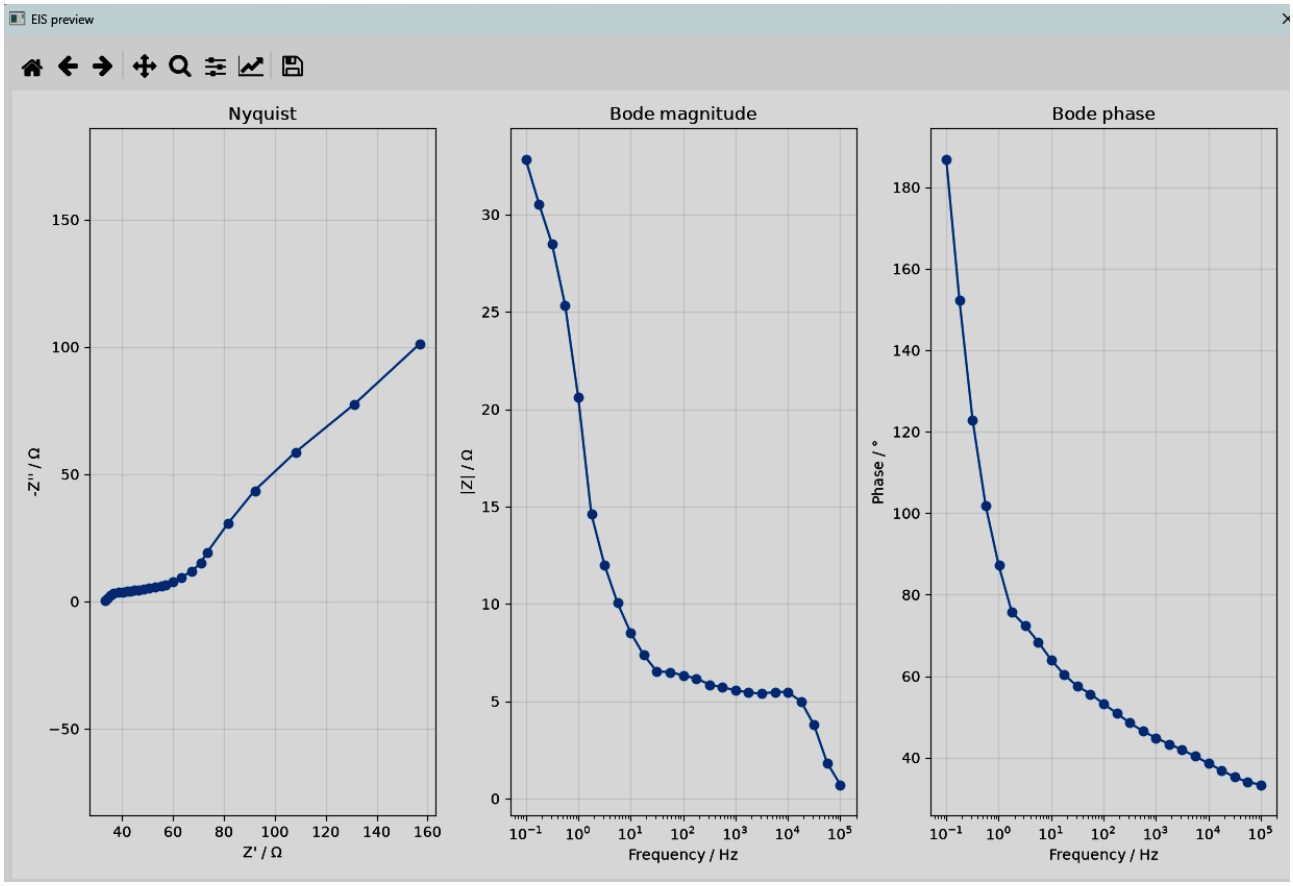


Figure 16: Preview of the imported impedance data before fitting, shown as a Nyquist plot (−Z'' versus Z') together with the Bode magnitude |Z| and the Bode phase φ as functions of frequency. Previewing the raw data makes it possible to identify outliers and incomplete frequency regions before an equivalent circuit is applied.

14.5 Randles-type equivalent-circuit fit

A simple Randles-type model contains the solution resistance R_s in series with an interfacial branch containing charge-transfer resistance R_ct and a capacitive element. For an ideal double-layer capacitor C_dl, the interfacial branch is R_ct in parallel with C_dl. More complex implementations may replace the ideal capacitor by a non-ideal capacitive element or include a diffusion contribution when justified by the data.
Equivalent-circuit parameters are phenomenological model parameters and should not automatically be assigned to a unique physical process. Select the simplest circuit that captures the experimentally observed features and is consistent with the electrochemical system.

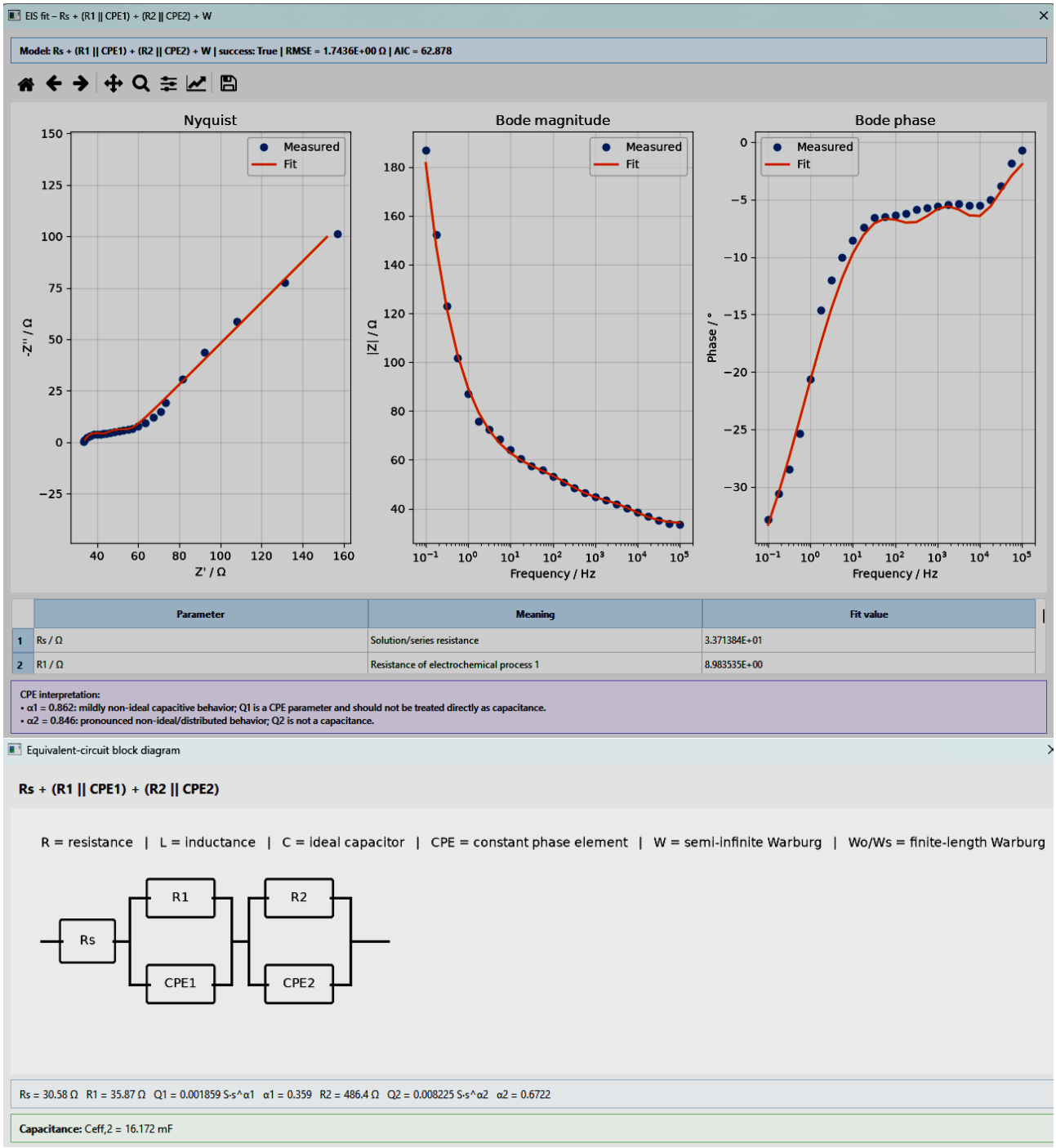


Figure 17: Result of the equivalent-circuit fit. Top: the measured data (points) and the fitted response (line) shown as Nyquist, Bode-magnitude and Bode-phase plots for the model Rs + (R1 ∥ CPE1) + (R2 ∥ CPE2) + W (here RMSE = 1.74 Ω, AIC = 62.9). Bottom: the block diagram of the fitted equivalent circuit with the corresponding parameter values.

14.6 Residual analysis

The module reports residuals between measured and fitted impedance values across the frequency range. Residual inspection is essential: a fit can appear visually satisfactory in the Nyquist plot while still showing systematic frequency-dependent errors. Randomly distributed small residuals support the selected model; structured residuals indicate that the model, weighting or data quality should be reconsidered.

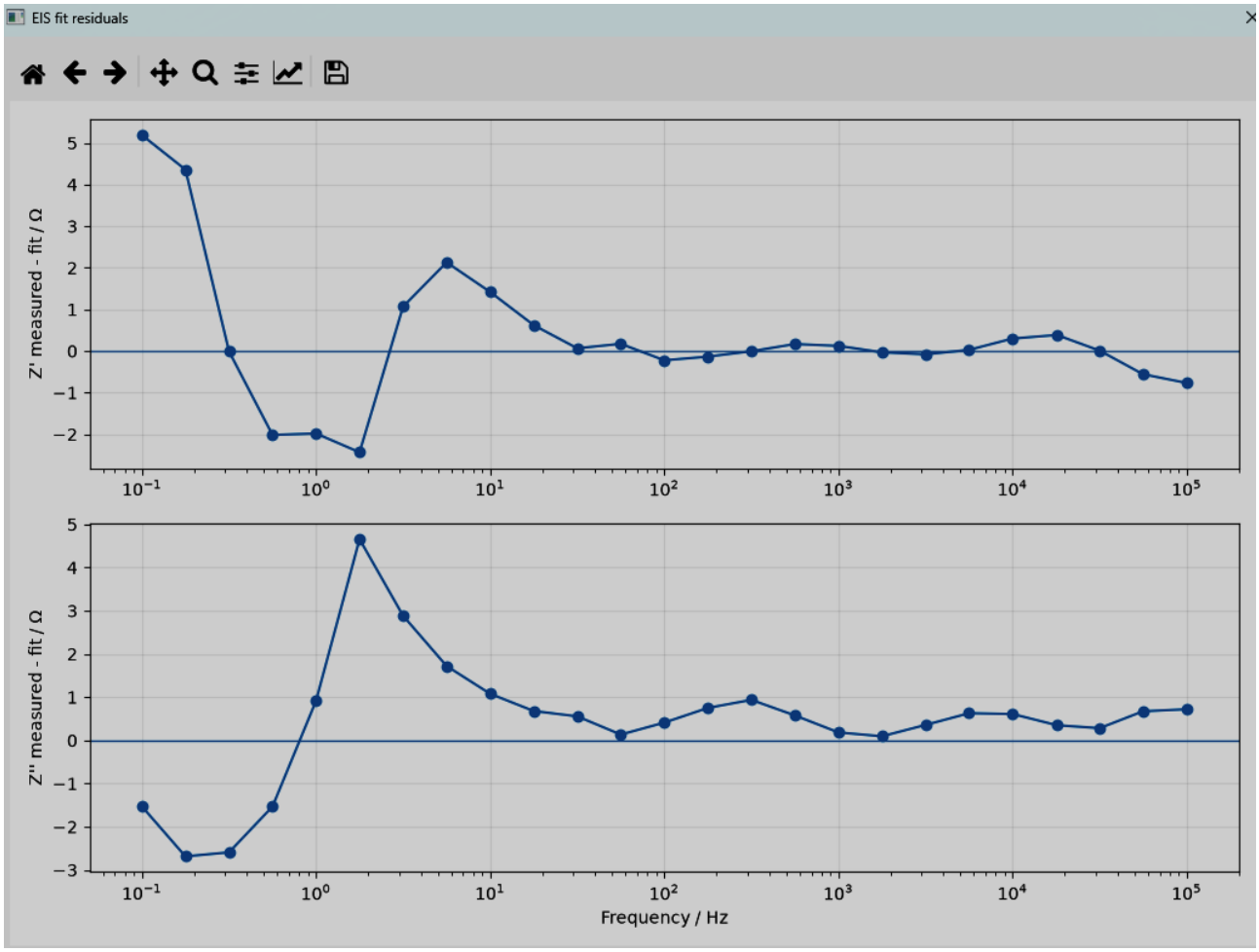


Figure 18: Residuals of the equivalent-circuit fit, shown as the difference between measured and fitted values for the real part Z' (top) and the imaginary part Z'' (bottom) as a function of frequency. Small, randomly distributed residuals support the selected model, whereas structured residuals indicate that the model, the weighting or the data quality should be reconsidered.

14.7 Practical fitting procedure

- Load the EIS CSV file and verify frequency and impedance columns.
- Inspect Nyquist and Bode plots before fitting and identify obvious outliers or incomplete frequency regions.
- Apply the available Randles-type circuit only when its elements are physically plausible for the experiment.
- Inspect fitted parameters and residuals over the complete frequency range.
- Repeat the fit with changed starting values only when necessary and check whether the same physically reasonable solution is recovered.

14.8 Outputs

- Nyquist plot of measured data and fitted response.
- Bode magnitude and phase plots.
- Fitted equivalent-circuit parameters.
- Residual plots or residual data for evaluation of fit quality.
- Excel export containing experimental data, fitted values, model parameters and residuals.

14.9 Scientific checks

Scientific check: Equivalent-circuit fitting is not unique. Parameter values are reliable only when the data span the relevant characteristic frequencies and the selected elements are identifiable. Large parameter correlations, implausible values or systematic residuals are evidence that the model should not be over-interpreted. Always report the circuit topology, frequency range and experimental conditions together with fitted parameters.

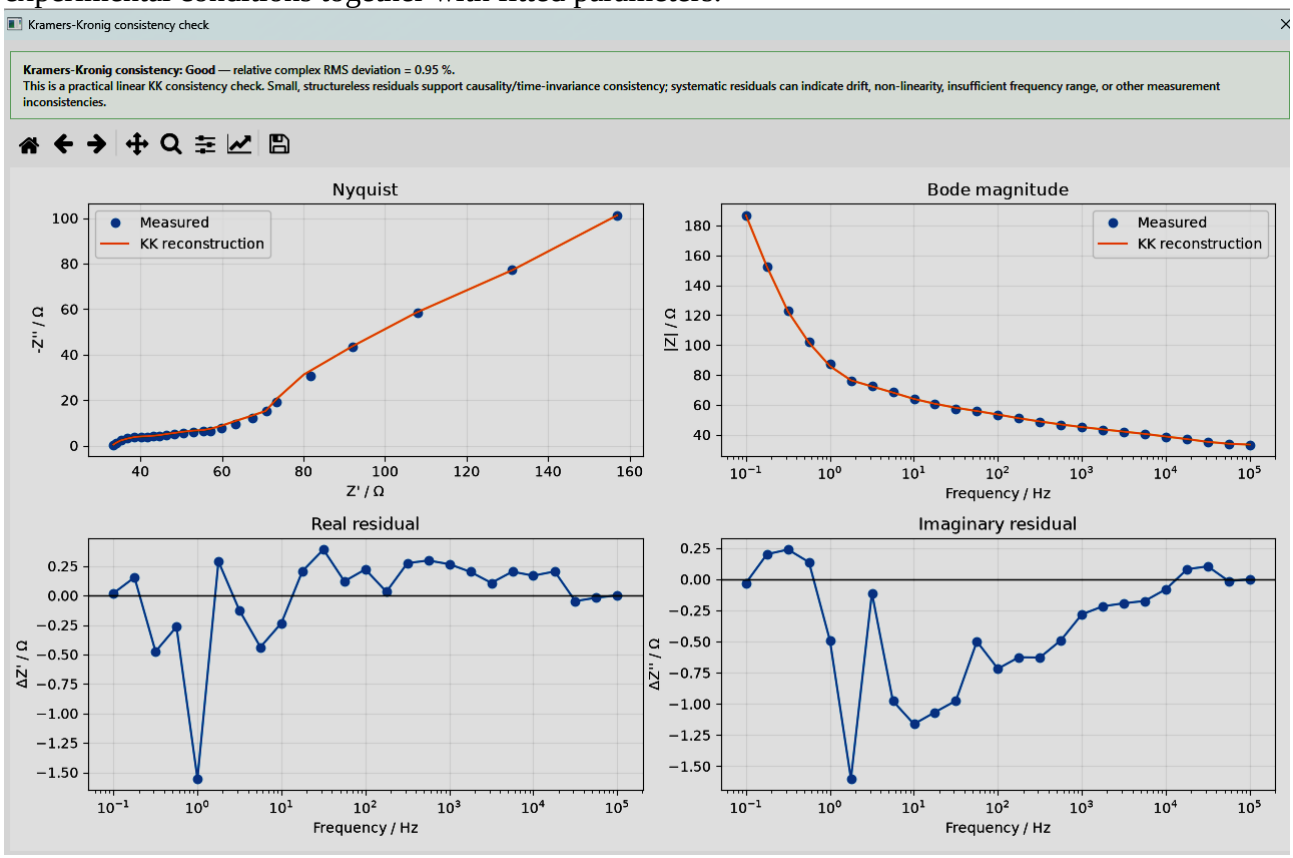


Figure 19: Kramers–Kronig consistency check. The measured data (points) and the KK reconstruction (line) are compared as Nyquist and Bode-magnitude plots, with the corresponding real and imaginary residuals shown below. A small, structureless deviation (here a relative complex RMS deviation of 0.95 %) supports causality and time-invariance, whereas systematic residuals can indicate drift, non-linearity or an insufficient frequency range.

15. Stripping Voltammetry

15.1 Purpose

The Stripping Voltammetry module is designed for quantitative evaluation of stripping-voltammetric concentration series recorded by square-wave voltammetry (SWV) and differential-pulse voltammetry (DPV). Both methods can be displayed and analyzed in parallel. The module combines voltammogram visualization, interactive selection of the integration interval, manual peak-current selection, concentration calibration and optional determination of an unknown concentration.

15.2 Input data

Load the SWV and/or DPV calibration CSV files. The first column contains the electrode potential and the first row contains the concentrations of the individual calibration measurements. Each subsequent column therefore represents one stripping voltammogram at a known concentration. Concentrations must be entered consistently, preferably in mol/L. The potential and current units used in the source files should be retained and documented.

15.3 SWV and DPV preview

The SWV and DPV voltammograms are shown in separate preview panels so that both methods can be inspected directly. The plots can be zoomed without losing manually selected points. A right mouse click restores the original plot range. The integration limits are represented by two red cursors; the two limits can be moved independently for SWV and DPV.

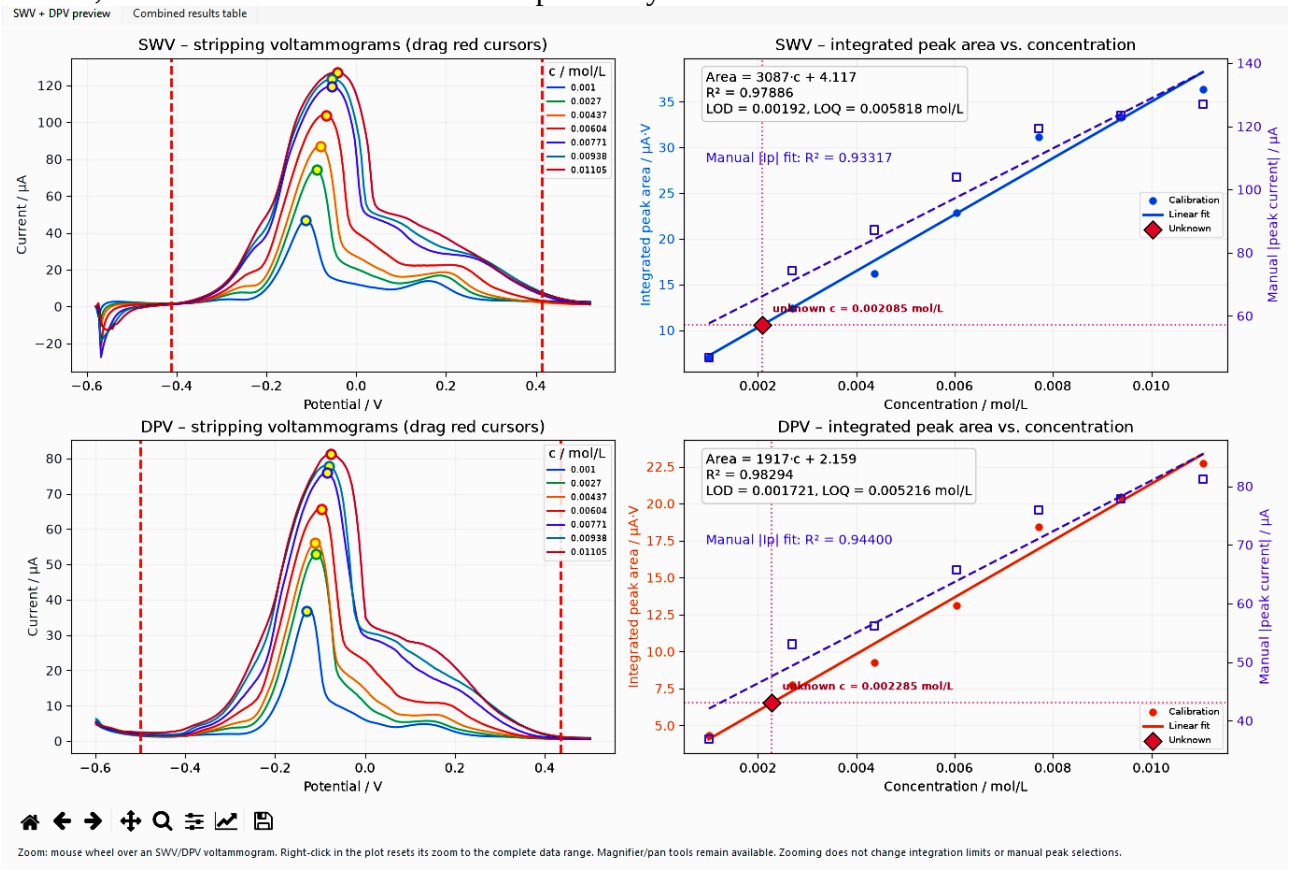


Figure 20: SWV and DPV measurements (left) and integral and peak values as a function of concentration (right). Red square: calculated unknown concentration.

15.4 Waterfall representation

Concentration-dependent stripping voltammograms can be displayed as waterfall plots. SWV and DPV data are retained as separate methods in the preview and in the exported plots. Interactive waterfall representations can be rotated and zoomed, allowing changes in peak position, peak shape and signal magnitude across the concentration series to be inspected.

15.5 Integrated peak area

For quantitative area evaluation, define the integration interval with the two movable cursors. The program numerically integrates the current within the selected potential interval for each

concentration. The same integration concept is applied to the unknown sample when one is loaded. The integration limits should include the complete analytical peak while excluding unrelated neighbouring processes as far as possible.

15.6 Manual peak-current selection

Peak currents are determined independently of the integration limits. The user selects the analytical peak manually by clicking the corresponding point on each voltammogram. The peak-current calibration curve is generated only after peak points have been selected. Moving the integration cursors therefore changes the integrated areas but must not change the manually selected peak currents.

15.7 Concentration calibration

The module provides concentration-dependent evaluation using either integrated peak area or manually selected peak current. The analytical response is plotted as a function of concentration and a calibration relationship is calculated from the known standards. Inspect the calibration graph and the individual voltammograms before accepting the regression; a high coefficient of determination alone does not demonstrate that the selected concentration range is analytically valid.

15.8 Determination of an unknown concentration

An optional CSV file containing a sample of unknown concentration can be loaded. Its integrated response is calculated using the same integration limits as the calibration series and entered into the integral-versus-concentration calibration. The corresponding concentration is calculated from the calibration function and displayed graphically. The unknown should be measured under the same electrode, electrolyte, deposition and pulse conditions as the standards and should lie within the calibrated response range whenever possible.

15.9 Measurement procedure documentation

The optional Measurement procedure section records experimental metadata without changing the numerical analysis. Common entries include analyte, working electrode, supporting electrolyte, reference electrode, deposition potential, deposition time and equilibration/rest time. Separate SWV and DPV tabs can be used to document method-specific pulse parameters, while an additional notes field can record further experimental details. These entries are documentation fields and do not alter peak integration or calibration.

15.10 Outputs

• SWV and DPV preview plots with independently movable integration cursors.
• Concentration-dependent waterfall representations, including interactive zoomable and rotatable views where enabled.
• Integrated peak area as a function of concentration.
• Manually selected peak current as a function of concentration.
• Graphical and numerical determination of an optional unknown concentration.
• PNG figures and Excel exports containing the voltammograms, selected limits, peak/area results, calibration data and documented measurement parameters; interactive HTML output is provided where enabled.

15.11 Scientific checks

Scientific check: Quantitative stripping voltammetry is particularly sensitive to electrode-surface condition, deposition potential and time, equilibration time, supporting electrolyte, pulse parameters and matrix composition. Standards and unknown samples must therefore be measured under comparable conditions. Verify that the selected integration interval contains the same analytical feature throughout the concentration series, that manually selected peak points correspond to the same process, and that the unknown is not evaluated by unjustified extrapolation outside the calibration range.

16. Electrochemical Surface Activation & SERS Analysis

16.1 Purpose

This module characterizes changes in Au and Ag electrode surfaces produced by electrochemical activation. It compares Raman spectra recorded before and after activation and can evaluate the corresponding cyclic voltammograms. The principal aim is to quantify how surface treatment modifies the Raman response under otherwise comparable experimental conditions.

16.2 Experimental concept

A typical experiment consists of four steps: (1) record a Raman spectrum of the untreated Au or Ag surface in the selected Raman-probe solution; (2) perform the electrochemical activation, for example in aqueous KCl; (3) rinse the electrode reproducibly; and (4) repeat the Raman measurement under the same optical conditions. Optional CV files recorded before and after activation can be loaded for both Au and Ag.

16.3 Raman input and x-axis conversion

The Raman x-axis can be supplied as Raman shift (cm^{-1}) or as measured wavelength (nm). When wavelength is selected, the laser wavelength is required and the software automatically converts the spectral axis to Raman shift. The default laser wavelength is 785 nm but can be changed. When the input already contains Raman shifts, the laser wavelength is retained as experimental metadata and is not required for the axis calculation.

16.4 Raman processing and baseline correction

The default baseline method is arPLS. The current default baseline parameter λ is 1000. Alternative methods include AsLS, polynomial baseline correction, edge-linear correction and no baseline correction. Savitzky–Golay smoothing can be enabled independently. Baseline-corrected spectra should always be inspected visually to ensure that broad Raman features are not removed or distorted.

16.5 Before/after comparison

For each metal, the program displays the baseline-corrected Raman spectra before and after activation as an overlay and calculates the difference spectrum (after − before). This representation makes newly enhanced, attenuated or shifted bands directly visible. The Raman display range can be restricted without changing the underlying imported data.

16.6 Apparent Raman enhancement

The program reports a global apparent Raman enhancement from the ratio of the integrated positive Raman areas after and before activation. Peak-specific apparent enhancement values are calculated from the corresponding peak intensities. The term apparent Raman enhancement is used deliberately: these ratios are not classical SERS enhancement factors unless the number of molecules contributing to the normal Raman and SERS signals is independently known.

16.7 Peak-specific analysis

Target Raman peaks may be entered manually, or the program can use automatic peak selection. For each selected band, the software compares the peak position and intensity before and after activation and reports the peak shift Δν together with the apparent intensity enhancement. The peak-search interval should be chosen according to the spectral resolution and expected band shifts.

16.8 Activation CV analysis

Four optional CV inputs are available: Au before activation, Au after activation, Ag before activation and Ag after activation. The before/after curves are overlaid for direct comparison. Individual cycles can be shown together with a mean representation. The program also evaluates activation-related charge quantities by integrating the current along the potential trajectory so that forward and reverse scans do not simply cancel.

16.9 Graphical interaction

The Raman/CV overview combines Raman overlays, Raman difference spectra and the corresponding CVs. Additional tabs show baseline control, enhancement results, activation CVs and the numerical summary. The plot area can be scrolled with the mouse wheel. The Matplotlib zoom tool allows rectangular zooming; the active zoom tool is highlighted, and RESET ZOOM restores the original view. The x-axes contain major and minor tick marks for more detailed reading.

16.10 Metadata

The experiment metadata include the Raman probe, measurement electrolyte, activation procedure, laser wavelength and probe concentration. These entries document the experimental context and are included in the exported results where applicable.

16.11 Outputs

• Raman before/after overlays for Au and Ag.

• Difference spectra (after − before).

• Global and peak-specific apparent Raman enhancement values.

• Peak positions and peak shifts.

• Before/after activation CV comparisons and charge-related quantities, when CV files are supplied.

• Excel export of processed spectra, numerical results and metadata.

• PNG export of the currently displayed plot.

16.12 Scientific checks

Scientific check: Enhancement ratios are meaningful only when the compared spectra were recorded under sufficiently similar optical and sampling conditions. Laser power, integration time, number of averages, focus, collection geometry, probe concentration and electrode positioning should therefore be kept constant. Surface activation may also change adsorption, roughness and local chemistry; an observed intensity increase should not automatically be attributed to electromagnetic SERS enhancement alone.



Figure 21: Module 10 overview showing Raman spectra before/after activation, Raman difference spectra and the corresponding Au and Ag cyclic voltammograms.

17. Signal-processing options

Signal processing is intended to improve data quality while preserving chemically meaningful spectral information. Excessive smoothing or baseline correction may distort quantitative results.

17.1 No processing

Use unprocessed data when the signal quality is sufficient or when the purpose is to document the original measurement. Raw outputs should be retained even when processing is applied.

17.2 Moving average

A moving average replaces each point by a local mean. It is simple and robust but can broaden peaks and shift apparent edges. It is best used for exploratory visualization rather than for quantitative peak-shape analysis.

17.3 Savitzky–Golay filter

The filter preserves polynomial features better than a simple average, but its behaviour depends strongly on the relationship between window length, polynomial order, sampling interval and peak width.

17.4 Baseline subtraction

Baseline methods estimate a low-frequency component. Edge behaviour and broad features require particular care. Record all baseline parameters in reports and retain the calculated baseline in the exported workbook.

17.5 Intensity clipping

Percentile clipping limits extreme values to improve color-scale usage. It should be reported explicitly when figures are used in publications, because it suppresses the displayed magnitude of outliers.

17.6 Logarithmic scaling

log1p scaling increases the visibility of weak features while compressing strong peaks. If negative values are shifted before the transformation, the transformed scale is no longer directly comparable to the original intensity scale.

18. Graphical outputs

18.1 3D surface

The surface plot maps potential, spectral coordinate and response value onto three axes. Rotate and zoom to inspect trends, but confirm interpretations with 2D views, because perspective can hide or exaggerate features.

18.2 Heatmap

The heatmap is often the clearest representation of potential-dependent band evolution. Inspect the color-bar range and check whether clipping or logarithmic scaling has been applied.

18.3 Contour plot

Contours emphasize lines of constant response and can reveal band shifts or transitions. Too many contour levels can suggest a precision that is not present in the measurement.

18.4 Waterfall plot

Waterfall plots show individual spectral traces. They are useful for following peak growth, decay and shift, but the applied offset must be stated or documented.

18.5 Interactive HTML

HTML files open in a web browser. The Plotly controls allow rotation, zooming, panning and image export. Some files may load the Plotly JavaScript library from a content-delivery network; offline use may therefore depend on how the HTML file was generated.

19. Excel and file exports

The exported Excel workbooks are intended to provide full numerical traceability of all graphical results.

19.1 Output location

If an output folder is selected, all generated files are written there. Otherwise, many modules use the folder containing the source CSV file. Use a dedicated analysis folder to avoid mixing raw and processed data.

19.2 Metadata

Metadata sheets document the source file, the selected range, the matrix dimensions, the smoothing method and the scaling method. Keep this sheet with the numerical outputs when data are transferred to collaborators.

19.3 Matrix and long formats

- Matrix format is convenient for surface and heatmap creation.
- Long format contains one row per spectral-coordinate/potential pair and is convenient for statistical software and database import.

19.4 File naming

Generated file names commonly include the source stem, the selected spectral range, the smoothing method, the scaling method and the plot type. Do not rename files in a way that removes processing information unless the information is preserved elsewhere.

20. Recommended scientific workflow

- Archive the original instrument export as read-only raw data.
- Create a working copy in CSV format.
- Check the axis orientation, units, missing values and matrix dimensions.
- Run the analysis first with no smoothing and no clipping.
- Inspect the raw plots for spikes, discontinuities and import errors.
- Apply the least aggressive processing that adequately addresses the analytical problem.
- Inspect the baseline and smoothing control plots.

- Export all numerical stages, not only the final figure.
- Record the software version, DOI and all parameters in the laboratory notebook.
- For publication, state whether values were baseline-corrected, smoothed, clipped, logarithmically transformed or vertically offset.
- Archive both raw data and processed results.

21. Troubleshooting

Problem	Likely cause	Recommended action
Batch file opens and closes immediately	Python or a package is missing	Open Command Prompt in the program folder and run the batch file there to read the error message.
“python” is not recognized	Python is not on PATH	Reinstall Python with “Add Python to PATH” enabled, or edit the batch file to use the py launcher.
ModuleNotFoundError	Packages were installed into another Python environment	Run “python -m pip install -r requirements.txt” using the same interpreter that starts the program.
CSV file not accepted	Non-numeric cells, wrong delimiter, incorrect matrix orientation or encoding	Open the file in a text editor, verify the first row/column, and export it again as CSV UTF-8.
No data points in the selected range	Range lies outside the detected axis values	Use the range-check function and enter limits within the available interval.
Savitzky–Golay error	Window is even, too small or longer than the available data	Use an odd window larger than the polynomial order and smaller than the number of spectral points.
Plot dominated by a single spike	Cosmic ray or extreme outlier	Inspect the raw data; correct the raw measurement if scientifically justified, or use percentile clipping for visualization only.

Problem	Likely cause	Recommended action
Baseline removes real bands	Baseline parameters are too aggressive	Reduce the polynomial flexibility or penalty settings and inspect the baseline overlay.
Interactive HTML is blank offline	Plotly JavaScript is referenced online	Connect to the Internet, or regenerate the HTML with embedded Plotly JavaScript.
Excel file cannot be overwritten	The workbook is open in Excel	Close the workbook, or select a different output file name or folder.
Application does not start from another folder	An old batch file used a fixed path	Use the release batch file, which starts relative to its own location, and keep the project files together.
Memory Error during export	Very large matrices	Reduce the spectral range or process the dataset in smaller sections

If interactive HTML plots do not open automatically, verify that a default web browser is configured in Windows.

22. Reproducibility and data integrity

- Retain the original CSV file and the instrument-native file.
- Record the exact software version and the version-specific DOI.
- Record all processing parameters and the selected spectral ranges.
- Record all processing parameters together with the software version and DOI.
- Preserve the exported metadata, raw matrices and processed matrices.
- Do not infer quantitative concentrations from arbitrary intensity values without calibration.
- Use the same processing settings when comparing samples, unless a justified exception is documented.
- Verify that the exported figures correspond to the numerical workbook used in the publication.
- Archive the generated Excel workbook together with the corresponding HTML and PDF outputs.

> Version-specific citation: For analyses performed with version 6.0, cite DOI 10.5281/zenodo.21283231. The concept DOI 10.5281/zenodo.21283231 refers to the software project across versions and resolves to the latest archived release.

23. Citation and licensing

23.1 Recommended citation

Users are encouraged to cite both the software DOI and any associated scientific publications where appropriate.

Habekost, A. (2026). SpectroElectroChem Suite (Version 6.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21283231

23.2 Software availability statement

SpectroElectroChem Suite is open-source software for the analysis and visualization of Raman, SERS, absorption and fluorescence spectro-electrochemical data. The source code is available from the GitHub repository, and the version used in this work is permanently archived on Zenodo (DOI: 10.5281/zenodo.21283231).

23.3 License

The software is distributed under the MIT License. Users may use, copy, modify and redistribute the software subject to the license terms. The software is provided without warranty. Consult the LICENSE file included in the repository for the full legal text.

23.4 Acknowledgement

Parts of the source code were developed with the assistance of OpenAI ChatGPT and were subsequently reviewed, scientifically validated, modified and substantially extended by the author.

Appendix A. Templates

A.1 Raman/SERS voltammogram template

CV:

WE(1).Potential (V)	WE(1).Current (A)	
0.79772949	1.00E-05	1.88E-05
0.79193115	2.27E-05	1.56E-05
0.78765869	1.81E-05	1.32E-05
0.78216553	1.47E-05	1.11E-05
0.77758789	1.21E-05	9.42E-06
0.77209473	1.00E-05	7.97E-06

0.76690674	8.29E-06	6.75E-06
0.7623291	6.88E-06	5.70E-06

Spectrum:

500.118	0.00015564	0.00926954	-0.01336714	-0.01032389	-0.00904945
500.704	-0.00103377	-0.00468765	-0.01085997	-0.01448833	-0.01616549
501.289	-0.00690933	8.52E-05	-0.0023204	0.00256249	0.00070913
501.874	-0.00707519	-0.00204674	-0.00068605	0.01187046	0.00130661
502.459	-0.013787	-0.00212257	-0.02538626	-0.02393781	-0.00753081
503.044	-0.00844831	-0.01350718	-0.02657903	-0.00545635	-0.00593803
503.629	-0.01748798	-0.01517641	-0.00090502	0.00192462	0.00809273
504.214	0.00063973	-0.00098241	0.02263814	-0.00398498	-0.02383873
504.799	-0.02689295	0.00921353	-0.00439898	0.00177432	-0.02906286
505.384	-0.0028603	-0.00900432	0.01059443	-0.00153577	-0.00606301

A.2 Absorption/fluorescence voltammogram template

Absorption

	0	0.05	0.1	0.15	0.2	0.25
189.009	0.0005471	-0.0009532	0.00027663	0.00030297	-0.00143763	-0.00188284
189.61	-0.00033372	0.00100445	0.00075553	-0.00075267	-0.00161739	-0.00060836
190.212	0.00189962	0.00375823	0.00062466	0.00083306	-0.00058677	9.72E-06
190.813	0.00370399	0.00057376	-0.00093781	0.00237169	0.00125474	0.00124634
191.414	-0.0003012	-0.00246313	-0.00160004	0.00042378	-0.00176299	-0.00344964
192.016	-0.00113445	0.00043965	0.00113964	0.00261128	0.00025744	0.00015962
192.617	-0.00056689	0.0012529	0.00351375	0.00148397	0.00230962	0.00012734
193.218	-0.00141669	9.35E-05	-0.00158593	-0.00083823	-0.00122175	-0.0007223
193.82	-0.00105767	-0.00109277	-0.00271187	-0.00326607	0.00041648	-0.00215385
194.421	5.95E-06	-0.0001413	-0.00029198	-0.00152576	-0.00089348	-0.00212903
195.022	0.0048457	0.00251859	0.00422578	0.00248135	0.00313149	0.00021809

Fluoroescence

	0	-0.04	-0.08	-0.12	-0.16
189.009	3.42941819	2.56644713	3.66035186	4.18552086	4.01285186
189.61	1.34191178	0.42888959	1.94750451	2.34257117	1.46000451
190.212	1.21858268	1.16059458	2.80940268	2.16953759	2.48690268
190.813	0.83454501	2.02759695	3.02229864	2.05951363	3.62979864
191.414	2.14600022	2.77234717	3.48738724	3.55119225	4.75988724
192.016	2.22524092	2.16718463	3.49560625	3.07560478	3.54310625
192.617	-0.8204931	0.30225194	1.90521461	2.27681579	2.74271461

A.3 Raman spectrum template

43.284	189
46.305	226
49.323	206
52.34	313
55.354	185
58.367	276
61.378	184
64.388	226

A.4 Absorbance spectrum template

189.009	0.0005471
189.61	-0.00033372
190.212	0.00189962
190.813	0.00370399
191.414	-0.0003012
192.016	-0.00113445
192.617	-0.00056689
193.218	-0.00141669
193.82	-0.00105767
194.421	5.95E-06
195.022	0.0048457
300	0.120

A.5 Fluorescence spectrum template

189.009	3.42941819
189.61	1.34191178
190.212	1.21858268
190.813	0.83454501
191.414	2.14600022
192.016	2.22524092

A.6 RRDE

WE(1).Potential (V)	311.11		522.22		733.33	
0.79742432	0.00044229	2.91E-06	0.00033456	2.87E-06	0.00029862	2.93E-06
0.7925415	0.00041458	2.93E-06	0.00031415	2.88E-06	0.00028064	2.93E-06
0.78704834	0.00038812	2.93E-06	0.00029474	2.90E-06	0.00026358	2.95E-06
0.78186035	0.00036304	2.96E-06	0.00027628	2.90E-06	0.00024738	2.96E-06
0.77697754	0.00033926	2.95E-06	0.00025876	2.91E-06	0.000232	2.97E-06
0.77209473	0.00031625	2.96E-06	0.00024191	2.91E-06	0.00021719	2.96E-06
0.76660156	0.0002945	2.96E-06	0.00022595	2.91E-06	0.00020325	2.98E-06
0.76202393	0.00027371	2.98E-06	0.00021082	2.92E-06	0.00018991	3.00E-06
0.75714111	0.00025394	2.98E-06	0.00019638	2.93E-06	0.00017728	3.00E-06
0.75195313	0.00023523	2.99E-06	0.00018274	2.94E-06	0.00016528	3.00E-06

Disk Ring

A.7 Spectro-CV Synchronization

CV, two cycles

0.79772949	1.00E-05	1.88E-05
0.79193115	2.27E-05	1.56E-05
0.78765869	1.81E-05	1.32E-05
0.78216553	1.47E-05	1.11E-05
0.77758789	1.21E-05	9.42E-06
0.77209473	1.00E-05	7.97E-06
0.76690674	8.29E-06	6.75E-06
0.7623291	6.88E-06	5.70E-06
0.75744629	5.69E-06	4.81E-06
0.75164795	4.68E-06	4.03E-06
0.74801636	3.83E-06	3.35E-06
0.7428894	3.10E-06	2.76E-06
0.73797607	2.47E-06	2.24E-06

Absorptionspectrum

500.118	0.00015564	0.00926954	-0.01336714	-0.01032389	-0.00904945
500.704	-0.00103377	-0.00468765	-0.01085997	-0.01448833	-0.01616549
501.289	-0.00690933	8.52E-05	-0.0023204	0.00256249	-0.00070913
501.874	-0.00707519	-0.00204674	-0.00068605	-0.01187046	-0.00130661
502.459	-0.013787	-0.00212257	-0.02538626	-0.02393781	-0.00753081
503.044	-0.00844831	-0.01350718	-0.02657903	-0.00545635	-0.00593803

503.629	0.01748798	-0.01517641	-0.00090502	0.00192462	-0.00809273
504.214	0.00063973	-0.00098241	-0.02263814	-0.00398498	-0.02383873
504.799	-0.02689295	-0.00921353	-0.00439898	-0.00177432	-0.02906286
505.384	-0.0028603	-0.00900432	-0.01059443	-0.00153577	-0.00606301
505.969	-0.02403207	-0.01349092	-0.00951039	-0.01491585	-0.00529247

Spectrum Raman

401.621	92	97	77	99
404.429	83	58	111	67
407.235	116	169	163	157
410.04	108	120	134	153
412.843	84	115	137	137
415.645	132	142	129	138
418.445	100	102	97	142
421.244	110	124	108	103
424.04	160	173	180	169
426.835	130	109	159	169
429.629	102	146	164	148

A.9 Diffusion

Chronoamperometry

0.300001	-8.13E-05
0.4	-7.06E-05
0.5	-6.33E-05
0.6	-5.80E-05
0.7	-5.42E-05
0.8	-5.08E-05
0.9	-4.79E-05
1	-4.56E-05
1.1	-4.37E-05
1.2	-4.20E-05
1.3	-4.05E-05
1.4	-3.92E-05
1.5	-3.80E-05
1.6	-3.70E-05
1.7	-3.60E-05

Chronocoulometry

0.4	1.13E-05
0.5	1.41E-05
0.6	1.67E-05
0.7	1.94E-05
0.8	2.20E-05
0.9	2.46E-05
1	2.72E-05
1.1	2.97E-05
1.2	3.22E-05
1.3	3.47E-05

Randles-Sevcik

	0.02	0.04	0.06	0.08	0.1	0.2	0.5
0.40103149	5.07E+00	6.47E+00	7.81E+00	1.18E+01	1.38E+01	1.83E+01	4.00E+01
0.39898682	4.64E+00	4.55E+00	8.08E+00	9.40E+00	1.20E+01	1.13E+01	2.50E+01
0.39682007	4.36E+00	5.39E+00	7.64E+00	7.71E+00	1.10E+01	1.11E+01	1.66E+01
0.39480591	4.14E+00	3.85E+00	5.83E+00	8.71E+00	1.04E+01	7.84E+00	1.14E+01
0.39297485	3.95E+00	4.77E+00	6.88E+00	8.80E+00	9.85E+00	8.86E+00	6.63E+00
0.39102173	3.80E+00	3.44E+00	6.24E+00	7.30E+00	9.45E+00	6.24E+00	7.29E+00
0.38909912	3.66E+00	4.32E+00	5.07E+00	6.05E+00	9.10E+00	7.58E+00	4.56E+00
0.38699341	3.54E+00	3.18E+00	6.33E+00	6.81E+00	8.80E+00	5.31E+00	3.07E+00
0.38531494	3.41E+00	3.92E+00	5.25E+00	7.67E+00	8.50E+00	6.65E+00	1.66E+00
0.38327026	3.32E+00	3.00E+00	4.64E+00	6.35E+00	8.28E+00	4.54E+00	-6.25E-01

A.10 EIS

Nyquist

Z'	-Z''	Frequency (Hz)
33.397835	0.421163	100227.93
33.963394	1.092473	56171.09
35.197594	2.327935	31653.688
36.733181	3.200445	17991.578
38.436516	3.684599	10009.775
40.149952	3.850893	5628.91
41.757233	3.937244	3165.368
43.167019	4.120324	1780.017
44.626251	4.351006	1000.977
46.301941	4.631064	562.891
48.289719	4.958586	316.536
50.631187	5.480479	178.001
52.876442	5.861152	100.098

55.313889	6.305648	56.289
57.155235	6.540702	31.653
59.864857	7.751007	17.799
63.302593	9.487741	10.01
67.418869	11.940914	5.628
70.829826	15.064724	3.165
73.262901	19.109241	1.78
81.607513	30.674637	1.001
92.08522	43.591499	0.563
108.069489	58.604588	0.317
131.169479	77.411385	0.178
156.965408	101.20845	0.1

Bode

Frequency (Hz)	Phase	|Z|
100227.93	0.72249	33.40049
56171.09	1.842352	33.980961
31653.688	3.783976	35.274494
17991.578	4.979424	36.872341
10009.775	5.475752	38.612717
5628.91	5.478638	40.334206
3165.368	5.386431	41.942444
1780.017	5.452406	43.363216
1000.977	5.568668	44.83786
562.891	5.711658	46.532967
316.536	5.862816	48.543636
178.001	6.177823	50.926933
100.098	6.325197	53.200294
56.289	6.503504	55.672142
31.653	6.528386	57.528263
17.799	7.377334	60.364548
10.01	8.523995	64.009659
5.628	10.0438	68.468163
3.165	12.007259	72.414154
1.78	14.61881	75.714043
1.001	20.600239	87.182106
0.563	25.332039	101.881828
0.317	28.470364	122.937027
0.178	30.547537	152.308746
0.1	32.813187	186.765335

Appendix B. Output-file glossary

Output	Meaning
Raw_Intensity_Matrix	Imported numerical matrix after axis and empty-cell validation.
Smoothed_Intensity_Matrix	Matrix after the selected smoothing operation.
Plotted_Intensity_Matrix	Values actually used for the visual output, after optional clipping or logarithmic scaling.
Waterfall_Unshifted_Values	Processed spectra before vertical displacement.
Waterfall_Shifted_Values	Spectra after addition of the waterfall offsets.
Waterfall_Offsets	Numerical offset applied to each spectrum.
Metadata	Source, range, dimensions and processing settings.
*.html	Interactive browser-based plot.
*.png	Static raster figure.
*.pdf	Static vector or page figure, depending on the export method.
*.xlsx	Excel workbook with numerical data and metadata.

Appendix C. Quick-start checklist

- Download the official ZIP archive from Zenodo or GitHub.
- Extract the ZIP archive completely.
- Install Python 3.10 or newer.
- Run Install_required_Python_packages.bat once.
- Run Test_Installation.bat.
- Start the application with Start_SpectroElectroChem_Suite.bat.
- Alternatively, start the ready-to-run Windows executable (SpectroElectroChem_Suite.exe) or use the installed shortcut – no Python installation is required in that case.
- Select the correct analysis module.
- Load a correctly formatted CSV file.
- Choose a valid spectral range.
- Use conservative processing parameters.
- Run the analysis and inspect the control plots.
- Keep the raw, processed and metadata outputs together.
- Cite version 6.0 with DOI 10.5281/zenodo.21283231.

Appendix D. Typical RRDE Parameters

Parameter	Typical value
Collection efficiency	0.25
Rotation	500–3000 rpm
O_2 concentration	1 mmol/L
Potential range	0.7 V to -1.5 V

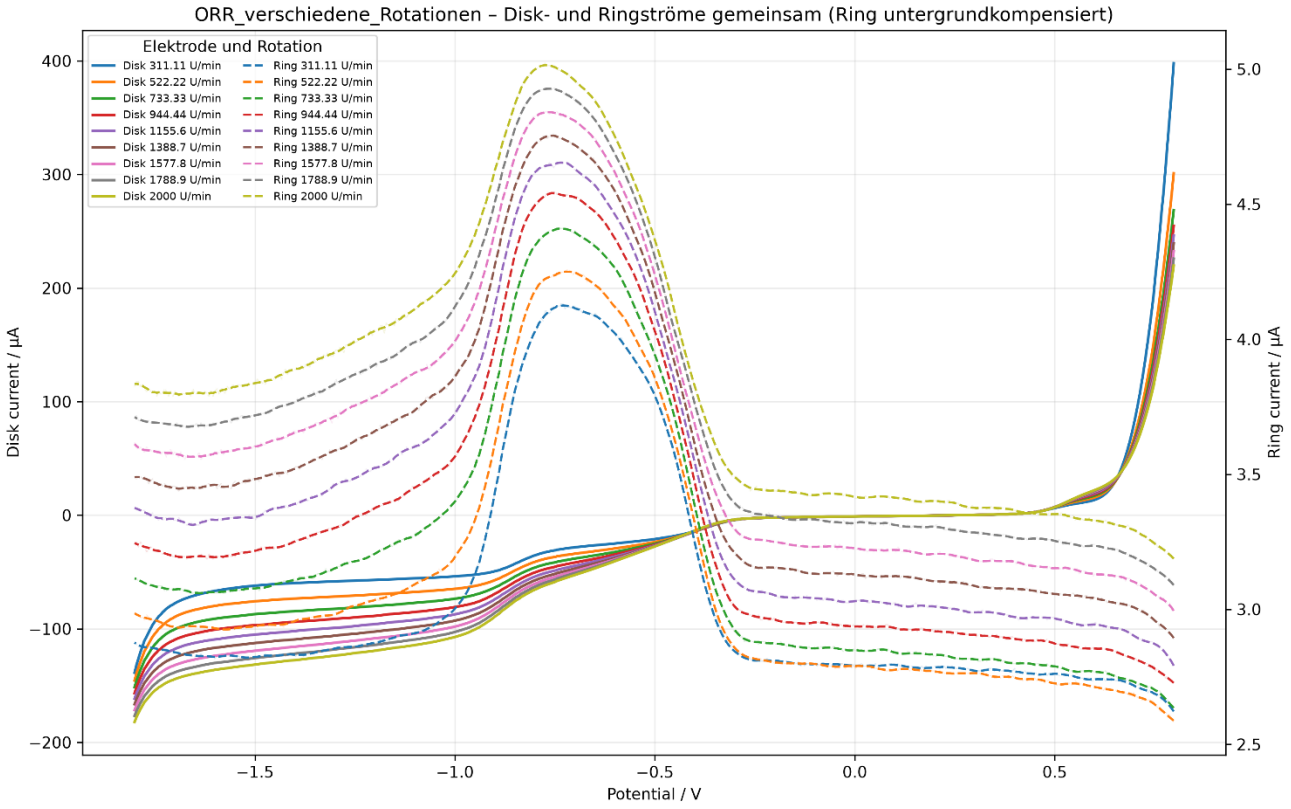


Figure 20: RRDE results for the oxygen reduction reaction (ORR): disk and ring currents between 0.7 and –1.8 V.

Appendix E. Installation from Python source (alternative route)

This appendix condenses the Python-source installation for advanced users and developers. Most users should instead use the executable or installer described in Section 4.5.

1. Install Python. Install Python 3.10 or newer and enable “Add Python to PATH” during setup.

2. Install packages. In the extracted program folder, run Install_required_Python_packages.bat (or py -m pip install -r requirements.txt).

3. Test. Run Test_Installation.bat to verify that Python and the principal dependencies import correctly.

4. Start. Launch the suite with Start_SpectroElectroChem_Suite.bat, or from a Command Prompt with python main.py.